# Multi-scale Walkability Measurement Framework :

## Application to Market Activity Analysis

**Ilho Jeong**, Ph.D. (Degree Expected, August 2026)
dlfgh4523@uos.ac.kr
Transportation planning lab, University of Seoul

Google Scholar

Linked in

# Ilho Jeong

## Ph.D. (August 2026)

Transport Planning Lab (TPL)

(2016 - 2022) ***Transportation Engineering and Data Science***, Bachelor's degree, University of Seoul

(2022 - 2026) ***Transportation Engineering and Transportation Planning***, Integrated Master's and Doctoral Program, University of Seoul

(2024 - 2024) ***Transport Studies Unit***, Visiting Ph.D. student**,** University of Oxford

## Journal papaers (7 articles in SCIE & SSCI, 3 articles in SCOPUS, 2 articles in KCI)

**1.** **Jeong, I.**, Choi, M., Kwak, J., Ku, D., & Lee, S. (2023). A comprehensive walkability evaluation system for promoting environmental benefits. *Scientific reports*, 13(1), 16183. **(SCIE, Q2)**

**2.** **Jeong, I.**, Chang, J., Lee, Y., Kwak, J., Kim, S., & Lee, S. (2026). Network Based Approach of Mobility Hubs for Sustainable Urban Mobility Systems. *In Proceedings of the Institution of Civil Engineers-Transport (pp. 1-18). Emerald Publishing Limited*. **(SCIE, Q3)**

3. Yu, Y., Kim, J., Choi, M., **Jeong, I**., Lee, S. (2025). Humanizing Transit Hubs: A GIS-based Approach to Boost Walkability in Gangnam. *Proceedings of the Institution of Civil Engineers: Municipal Engineer*. (**SCIE, Q4**)

4. Lee, Y., Bencekri, M., Kwak, J., **Jeong, I**., Lee, S. (2025). Electrification Pathways Towards Transport Decarbonization with a Systematic Review. *Clean Technologies and Environmental Policy*. (**SCIE, Q2**)

5. Kim, S., **Jeong, I.**, Choi, M., Kwak, J., Bencekri, M., & Lee, S. (2024). Fractal dimensional analysis to reveal traffic flow dynamics as organic metabolism. *Transportmetrica A: Transport Science*, 2390009. (**SSCI, Q3**)

6. Kwak, J., **Jeong, I.**, Lee, D., Ku, D., & Lee, S. (2024, August). Lessons learned from spatiotemporal effect of COVID on the population density in the CBD. *Proceedings of the Institution of Civil Engineers: Municipal Engineer* (Vol. 177, No. 4, pp. 169-179). Emerald Publishing Limited. (**SCIE, Q4**)

7. Lee, Y., Lee, S., Jeong, I., Kwak, J., Kim, S., & Lee, S. (2025). Reinforcement Learning-Based Planning of Dedicated Autonomous Vehicle Lanes for Adaptive Infrastructure Design. *Proceedings of the Institution of Civil Engineers: Municipal Engineer*. (**SCIE, Q4**)

**8.** **Jeong, I.**, Choi, M., Ku, D., & Lee, S. (2022). Activation plans of public transport based on urban network analysis. *Chemical Engineering Transactions*, 97, 169-174. (SCOPUS)

9. Jo, J., Kim, G., Kwak, J., Jeong, I., Ku, D., & Lee, S. (2023). Air quality modeling using real-time Urban big data. *Chemical Engineering Transactions*, 106, 229-234. (SCOPUS)

10. Kwak, J., Oh, H., **Jeong, I.**, Shin, S., Ku, D., & Lee, S. (2021). Changes in shared bicycle usage by COVID-19. *Chemical Engineering Transactions*, 89, 169-174. (SCOPUS)

11. Park, J., Ku, D., **Jeong, I.**, & Lee, S. (2022). A study of walking activity time characteristics based on the time use survey. *Journal of the Korea Institute of Intelligent Transportation Systems*, 21(3), 53-61. (KCI)

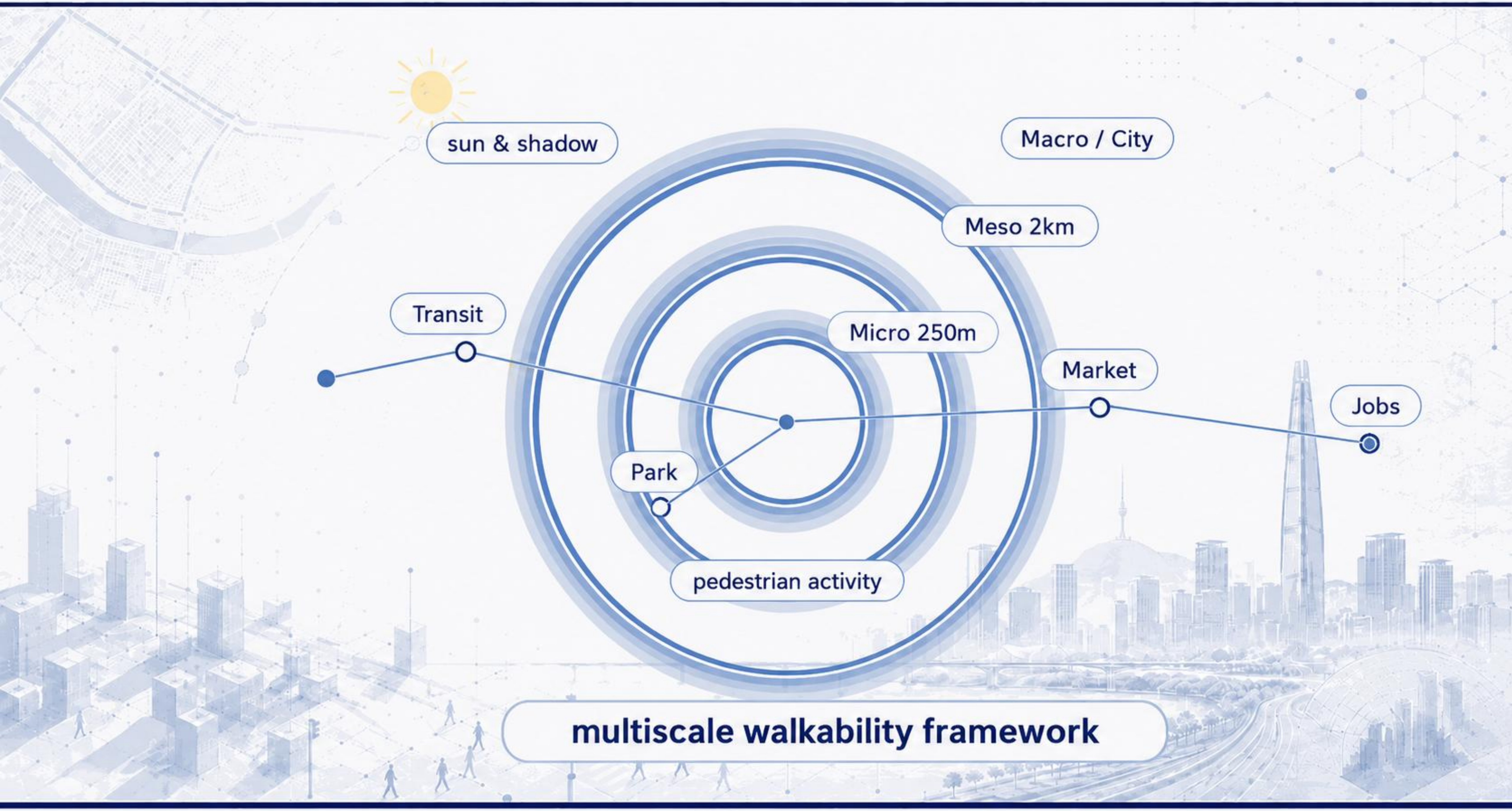
12. Kwak, J., Ku, D., Oh, H., **Jeong, I**., Shin, S., & Lee, S. (2022). Impact Analysis in Spatial Use Changes by COVID: Focusing on Shared Bicycles. *Journal of Korean Society of Transportation*, 40(2), 178-189. (KCI)

## Research in progress

**1.** **Jeong, I.**, Park, S., Lee, Y., Kwak, J, Kim, S., & Lee, S. (2026). A Pedestrian-Aware, Energy-Efficient Pathfinding Framework for Autonomous Delivery Robots. *Transportation Research Record*.

2. Kim, S., **Jeong, I.**, … , Lee, S. (2026). Prioritising Street Space Reallocation:A Spatial Framework for Urban Transport Planning. *International Journal of Urban Sciences.*

**3.** **Jeong, I.**, … , Lee, S. (2026). Urban Walkability and Mental Health:A Global Meta-analysis of Evidence across Cities. *Nature communications.*


sun & shadow
Macro / City
Meso 2km
Transit
Micro 250m
Market
Jobs
Park
pedestrian activity
multiscale walkability framework

# 1. Motivation

I. Walkable cities

II. What makes a city walkable?

III. Emerging challenges for walkability

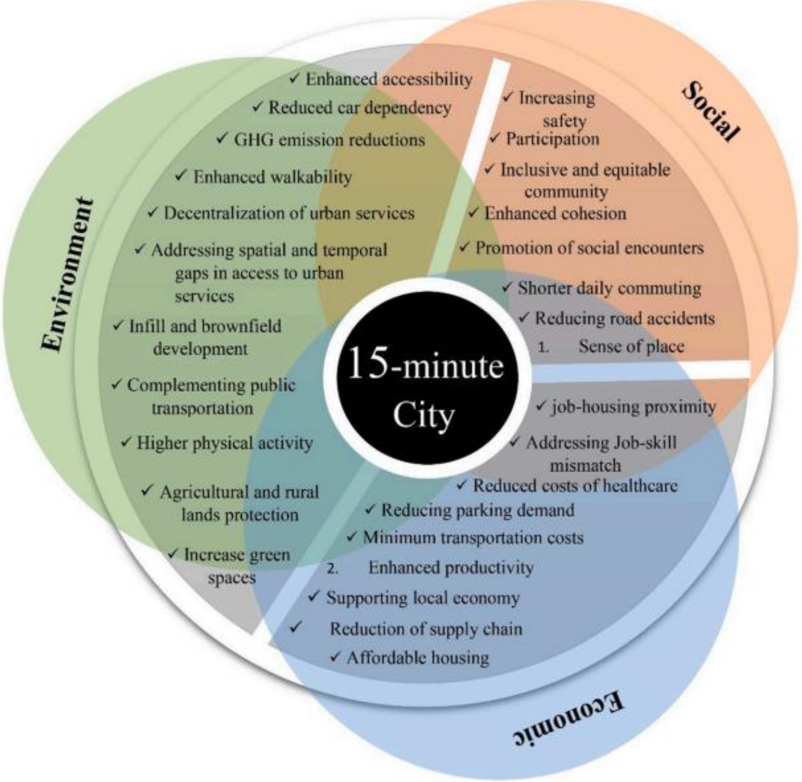
IV. Benefits of Walkability

# 01 Motivation : Walkable cities

- Many cities are promoting walkable urban environments as part of their urban and mobility planning.
- Their benefits extend beyond walking itself, contributing to environmental, social, and economic sustainability.

**Walkable urban environment**

**Walkable cities to urban sustainability**




S: Khavarian-Garmsir, A. R. et al., 2023

# 01 Motivation : What makes a city walkable?

- **Walkability is shaped by multiple factors operating at different spatial scales.**

# 01 Motivation : Emerging challenges for walkability

- **Climate change is making thermal comfort an increasingly important walking condition.**

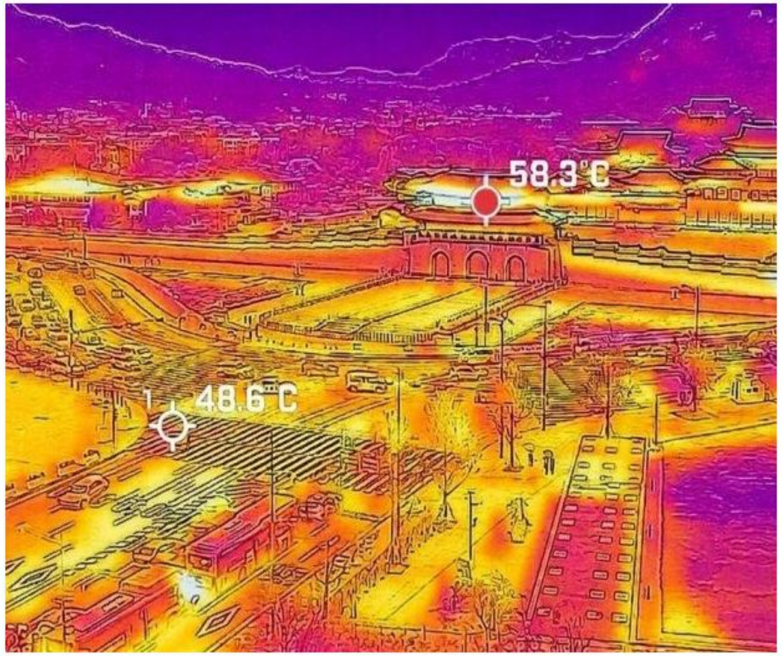
**Extreme Heat in Seoul**

**Shade-Seeking Behavior under Extreme Heat**

Seoul issues highest heat alert amid extreme temperatures

Posted August. 05, 2026 08:49, Updated August. 05, 2026 08:49

A− A+ 한국어



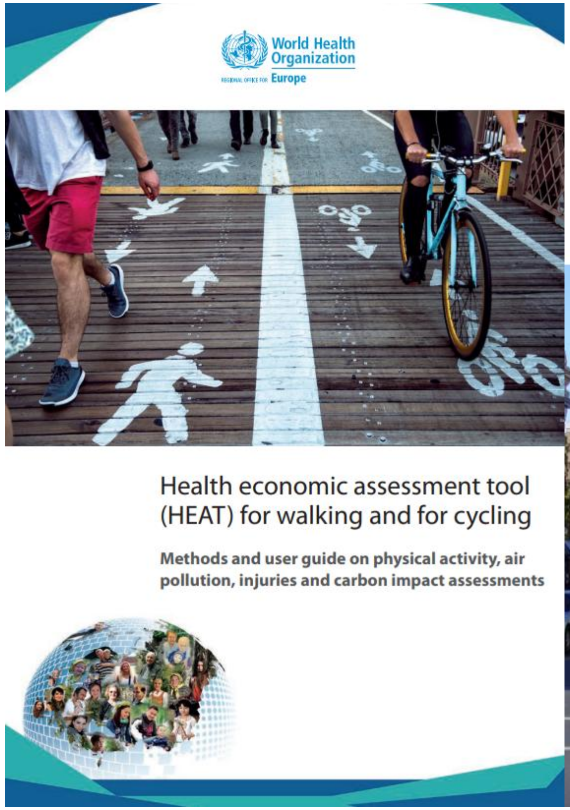

Headline News

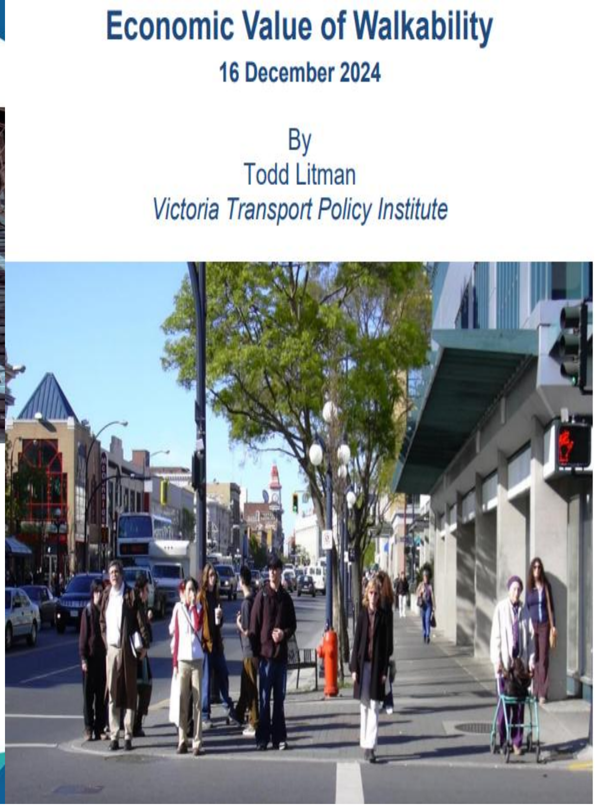
- Extreme heat pushes Seoul to 40 degrees
- Lee orders all-out climate disaster response
- Industry minister calls for work-hour reform
- Pentagon considers tactical nuclear strategy shift
- Kim Dae-won scores memorable goal against Man City

# 01 Motivation : Benefits of Walkability

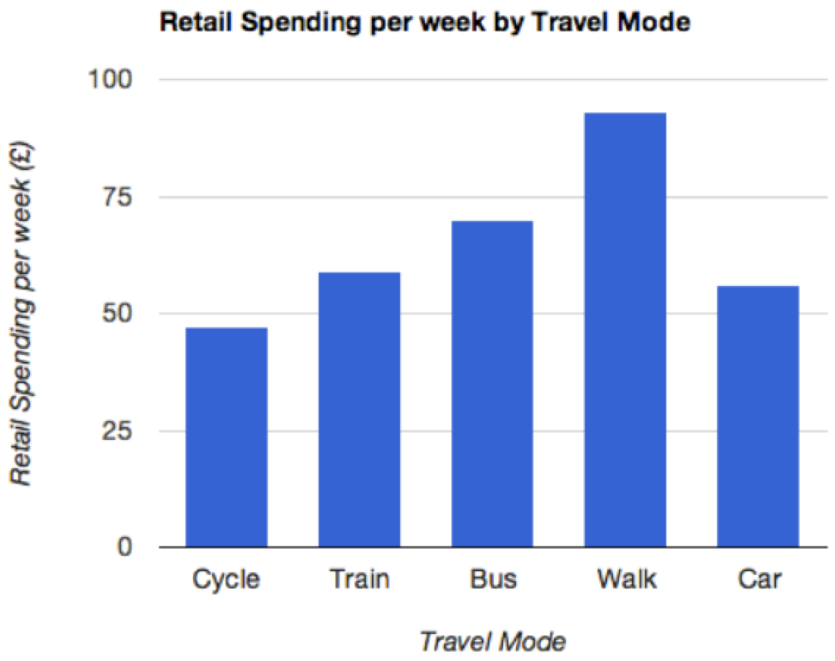
- **Walkable environments can reduce transport externalities while providing measurable health and economic benefits.**
- **Walkability can also create wider economic value through local business activity, property values, investment, and urban efficiency.**

## Health and Economic Benefits of Walking and Walkability



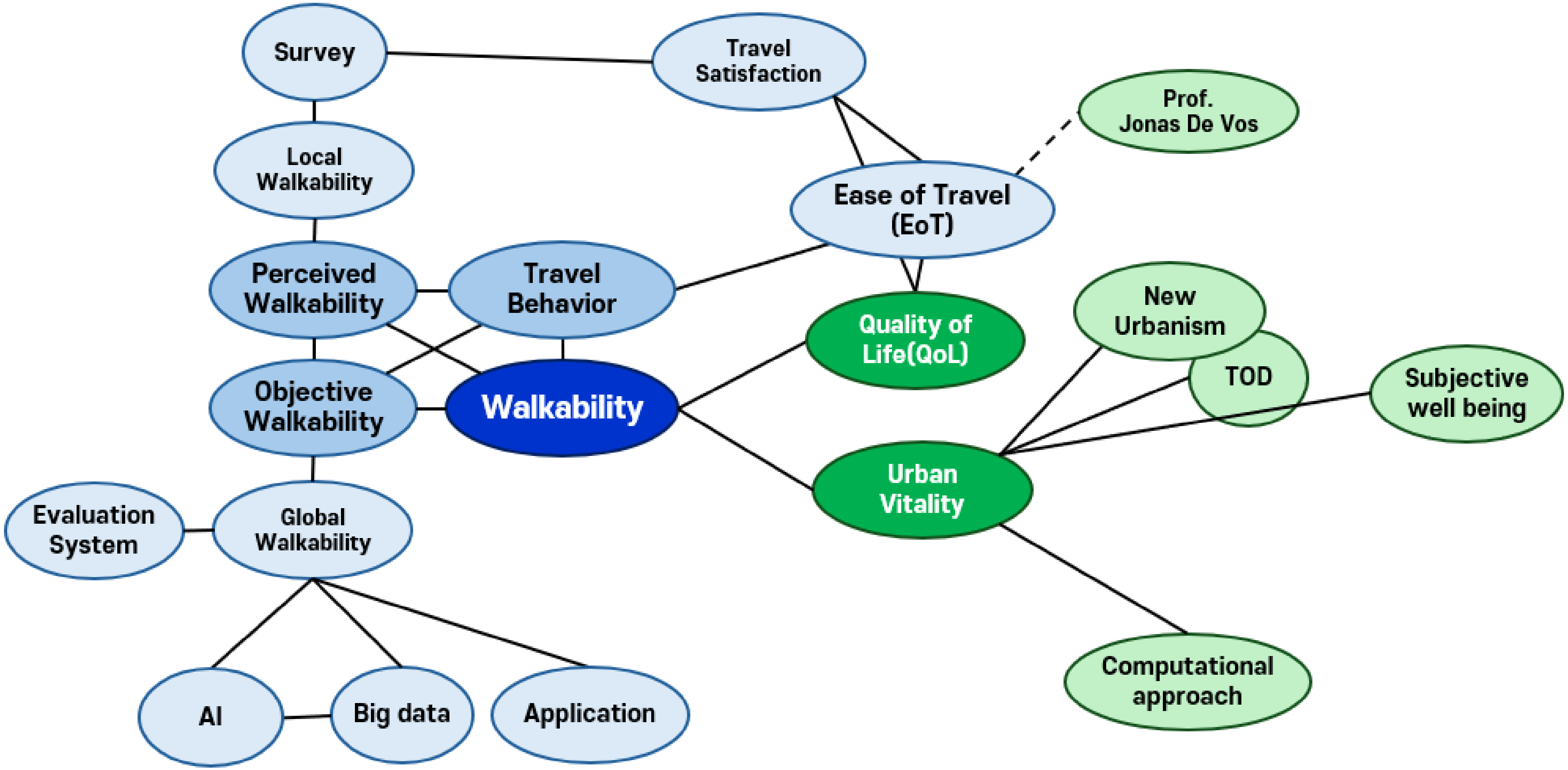




## Retail Spending by Travel Mode

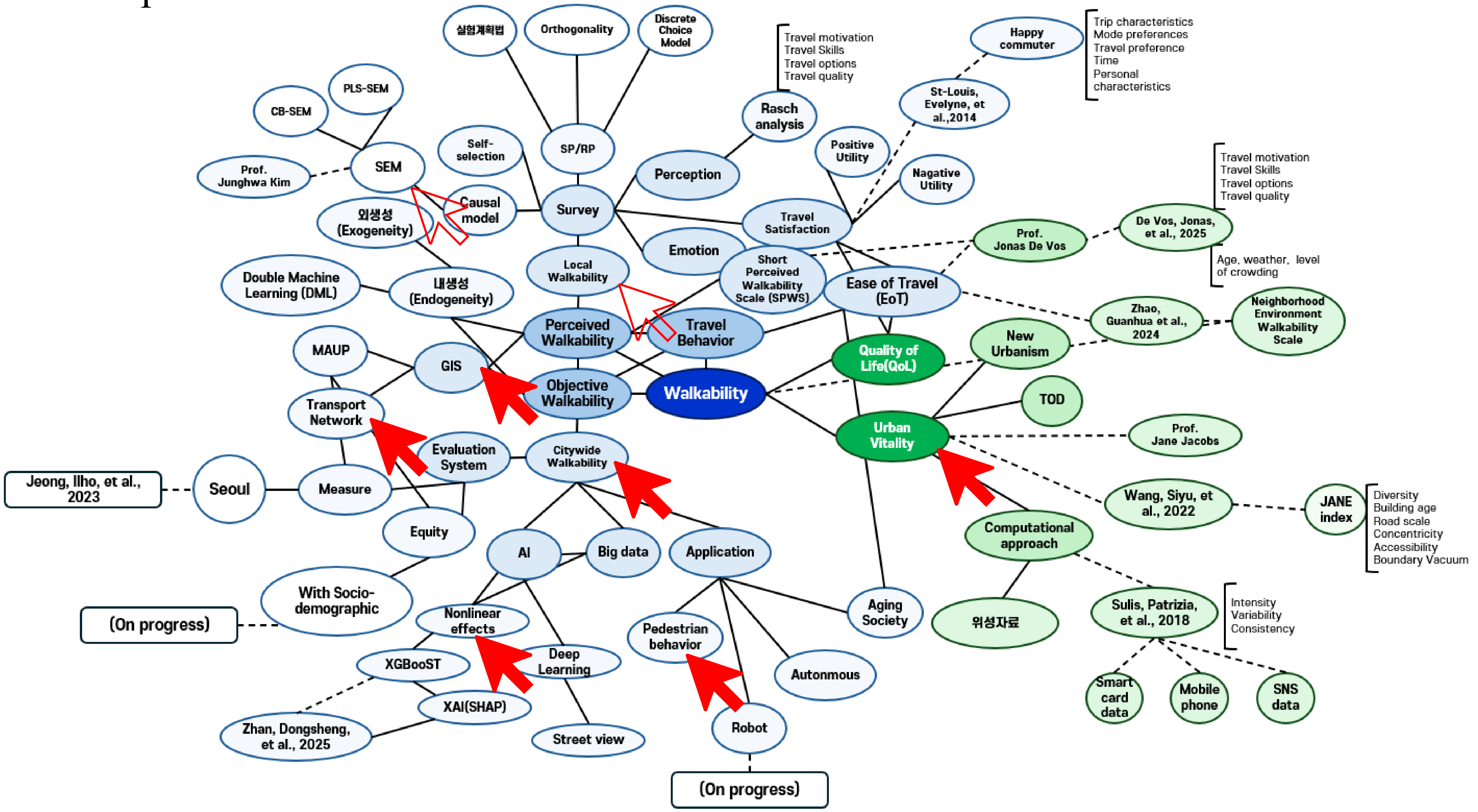



S: https://retail-insider.com/retail-insider

# Research map



# Research map



# 01 Research Gaps and Contributions

- **Research Gaps**

1) Walkability is shaped by spatially and temporally varying conditions that cannot be fully captured by a single, fixed indicator.
2) The effects of walkability on actual walking behavior may differ across spatial scales.
3) The influence of walkability may extend beyond walking behavior to broader urban outcomes, including commercial activity.

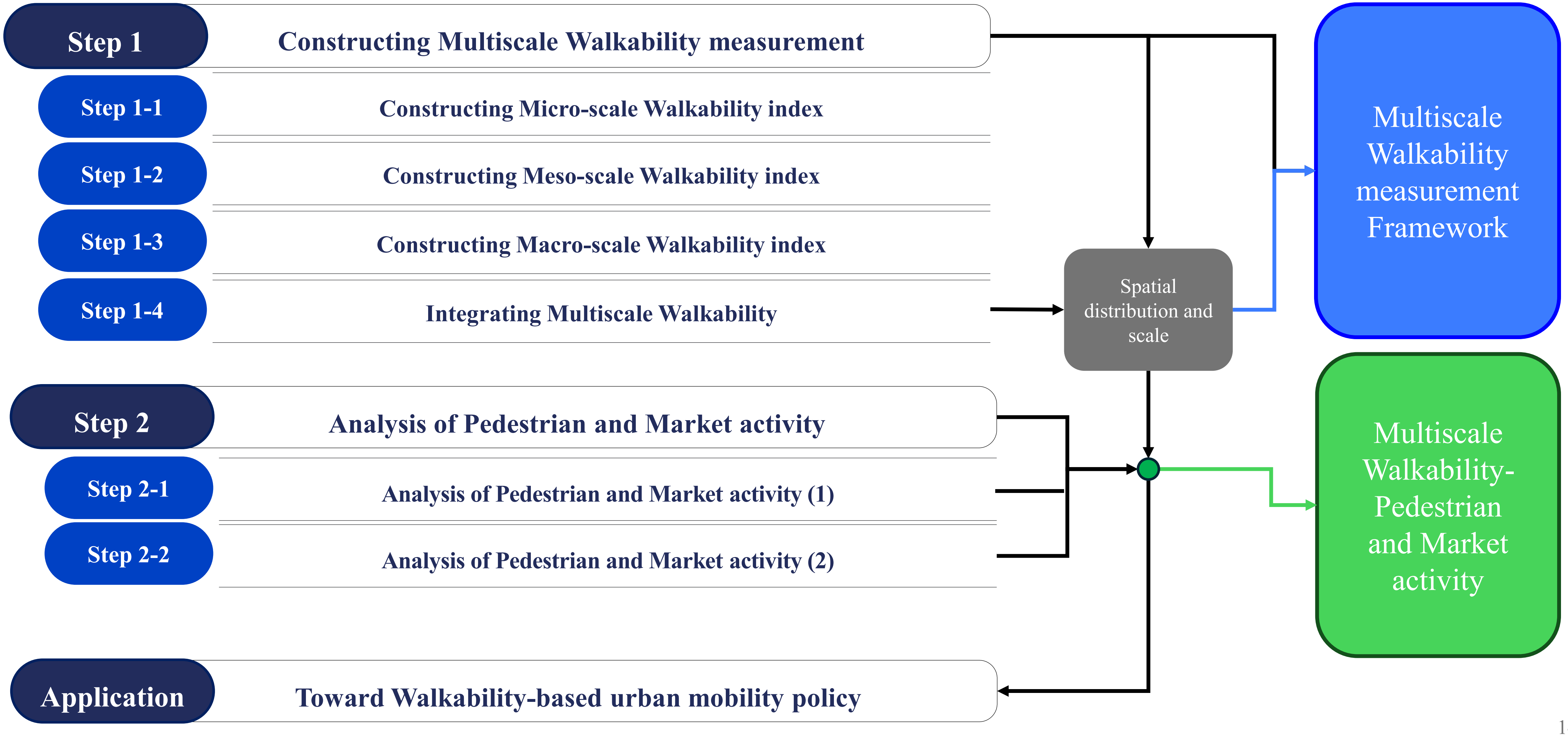
- **Research Contributions**

# 2. Research Design

I. Research Framework

II. Literature Review and Research Gap

# 02 Research Framework



# 02 Literature Review and Research Gap

## Literature Review

### 1. Walkability components
- Density, land-use mix, street connectivity, accessibility, and transit access.

### 2. Walking distance and exposure range
- Thresholds range from about 150 m to over 2 km, depending on purpose and context.

### 3. Scale-specific indicators
- Indicators have been developed separately for city/block, network, and street scales.

### 4. Seasonal and thermal conditions
- Shade and thermal comfort affect route choice and walking experience.

### 5. Walkability and commercial activity
- Walkability is linked to retail or business performance, usually at limited scales.

## Research Gaps

### 01 Limited multiscale perspective
Existing studies often examine walkability at a single spatial scale, with limited attention to how its effects differ across street-, neighborhood-, and city-scale conditions.

### 02 Limited consideration of behavioral range and temporal conditions
Walking distance, exposure range, and seasonal thermal conditions are often treated separately, despite their combined influence on actual walking behavior.

### 03 Limited evidence on downstream urban outcomes
How multiscale walkability translates into pedestrian activity and subsequently into commercial activity remains insufficiently understood.

# 3. Methodology

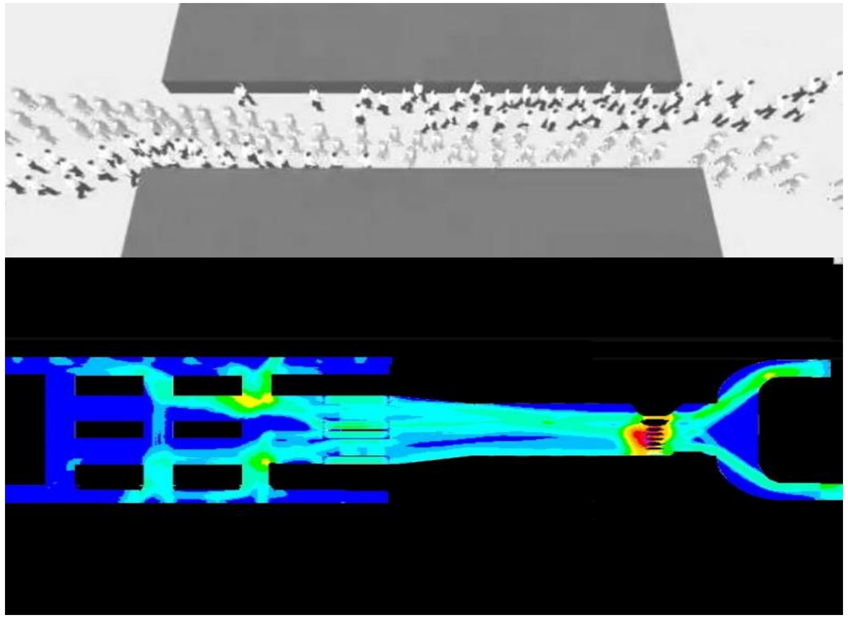
I. Multiscale Walkability Measurement

II. Study Scope and Data

# 03 Multiscale Walkability(1): Defining the Micro-Scale Walkability Index

- Conventional transportation studies have mainly evaluated walking through pedestrian flow and service levels, with limited consideration of spatial context.
- This study defines micro-scale walkability as local physical and network conditions within a 250 m grid.

**Examples of conventional pedestrian evaluation (Viswalk)**



**Micro-scale walkability index**

$$W_{micro} = Z(\ln(SW+1)) - Z(\ln(SL+1)) + Z(\ln(CW+1))$$

$W_{micro}$ =micro-scale walkability

SW = sidewalk length within the grid

$SL$ = average slope within the grid

$CW$ =number of crosswalks within the grid

**→ The logarithmic form : potential nonlinear effects of infrastructure supply.**

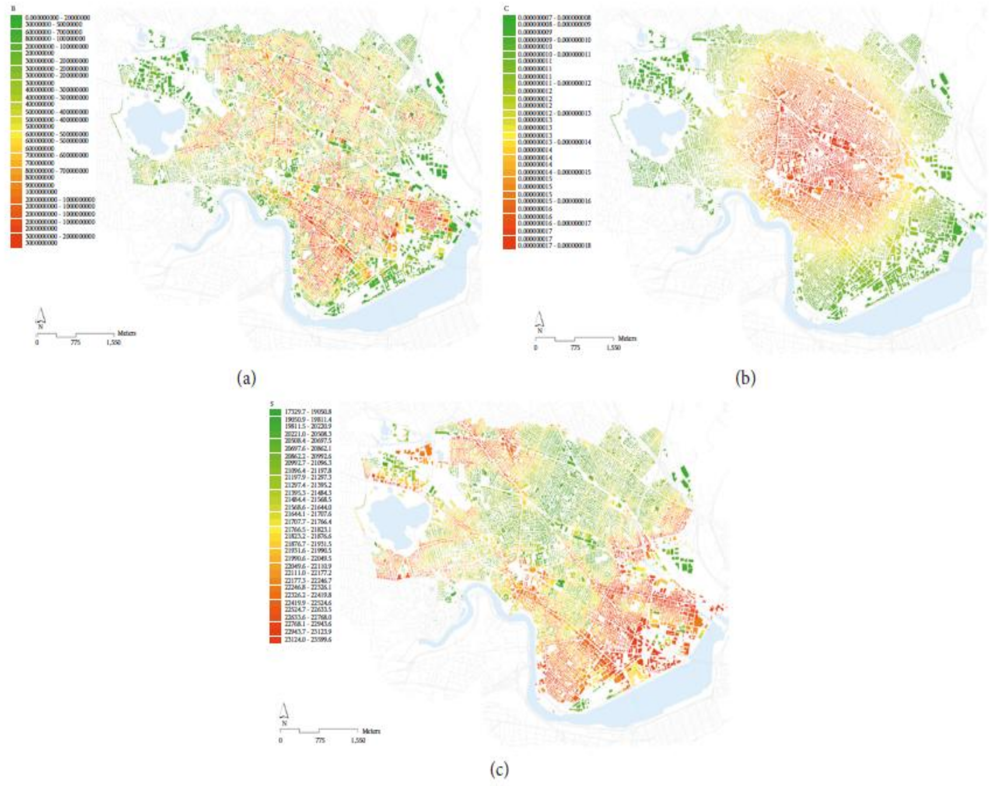
**→ Each indicator is standardized using Z-scores to ensure comparability.**

S: FRIIS, C., & SVENSSON, L. 2013.

# 03 Multiscale Walkability(2): Defining the Meso-Scale Walkability Index

- In this study, meso-scale walkability quantifies relative accessibility to public transport stations using Urban Network Analysis (UNA), capturing factors that influence pedestrian route choice.
- It also incorporates seasonal comfort by considering sun shadow frequency as a factor that can encourage or discourage walking.

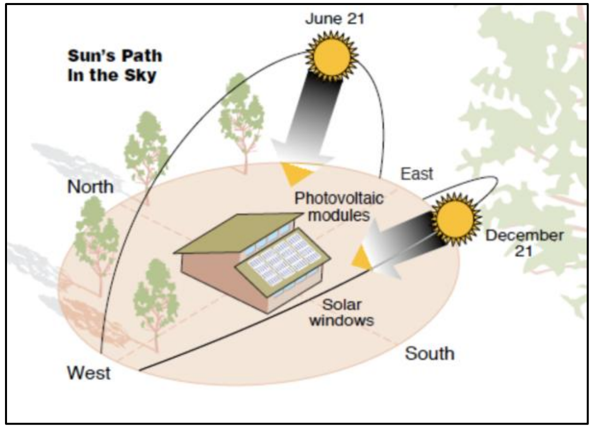
## UNA-Based Network Accessibility Analysis




S: Gallotti, R., Sacco, P., & De Domenico, M. 2021.

## Solar Shadow Frequency Analysis

Shadow length varies with seasonal solar altitude ($\alpha$), considering latitude, longitude, date, and time.




Shadow length is calculated based on 3D building height (H), and accumulated over time to estimate shadow exposure ($ST_i$).

$$L_{shadow} = \frac{H}{\tan(\alpha)}$$

$$ST_i = \sum_{t=1}^{T} S_i(t) \cdot \Delta t$$

$$S_i(t) = \begin{cases} 1 & if\ shaded \\ 0 & if\ sunlight \end{cases}$$

# 03 Multiscale Walkability(2): Defining the Meso-Scale Walkability Index

- **Meso-scale walkability links the micro scale (local physical conditions) with the macro scale (urban structure), focusing on route choice and connectivity within the pedestrian network.**
- **It represents walking-related opportunities and constraints within a 2 km range, including access to destinations, public transport accessibility, seasonal comfort, and development density (FAR).**

## Components of Meso-Scale Walkability

| | |
|---|---|
| **Destination accessibility** | **Average network distance to major POIs** |
| **Public transport accessibility** | **Accessibility to bike, bus, and subway stations** |
| **Seasonal comfort** | **Average seasonal shadow frequency** |
| **Development density (FAR)** | **Average floor area ratio** |

## Meso-Scale Walkability Index

$$W_{meso} = Z(\ln(A+1)) + Z(\ln(Bet * PT + 1)) + Z(\ln(SS+1)) + Z(\ln(FAR+1))$$

$W_{meso}$ = meso-scale walkability

A =number of major destinations (POIs)

$Bet$ =network centrality (betweenness)

$PT$ =public transport station accessibility

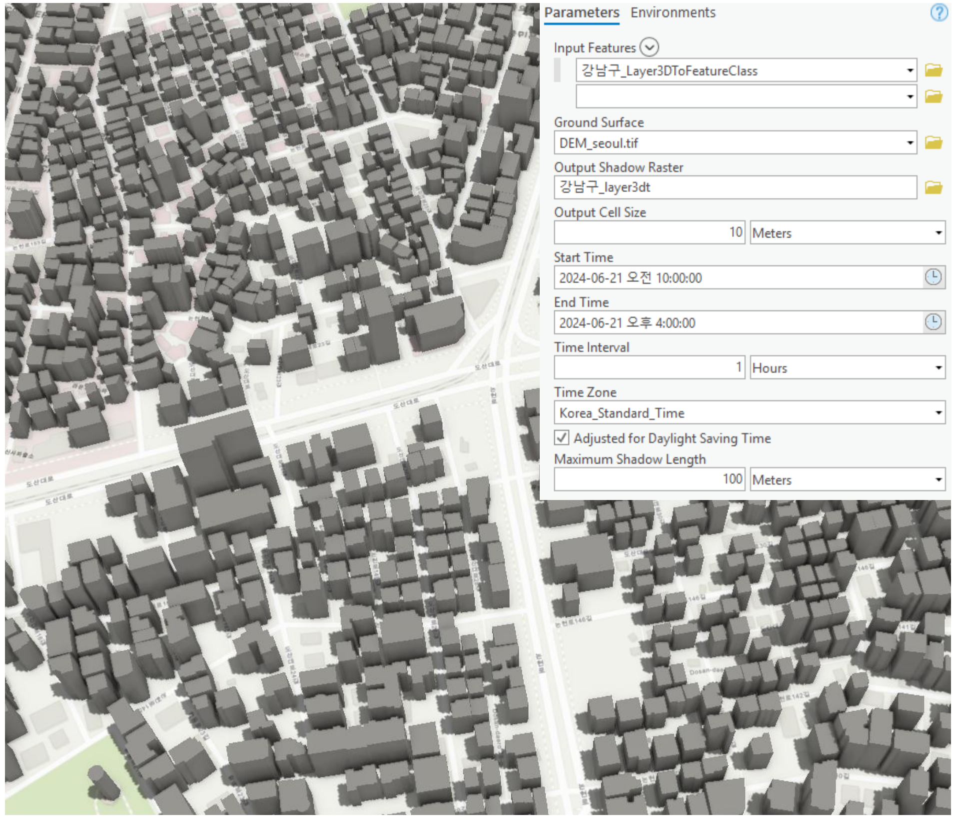
$SS$ =seasonal shadow frequency along sidewalks

FAR =average floor area ratio

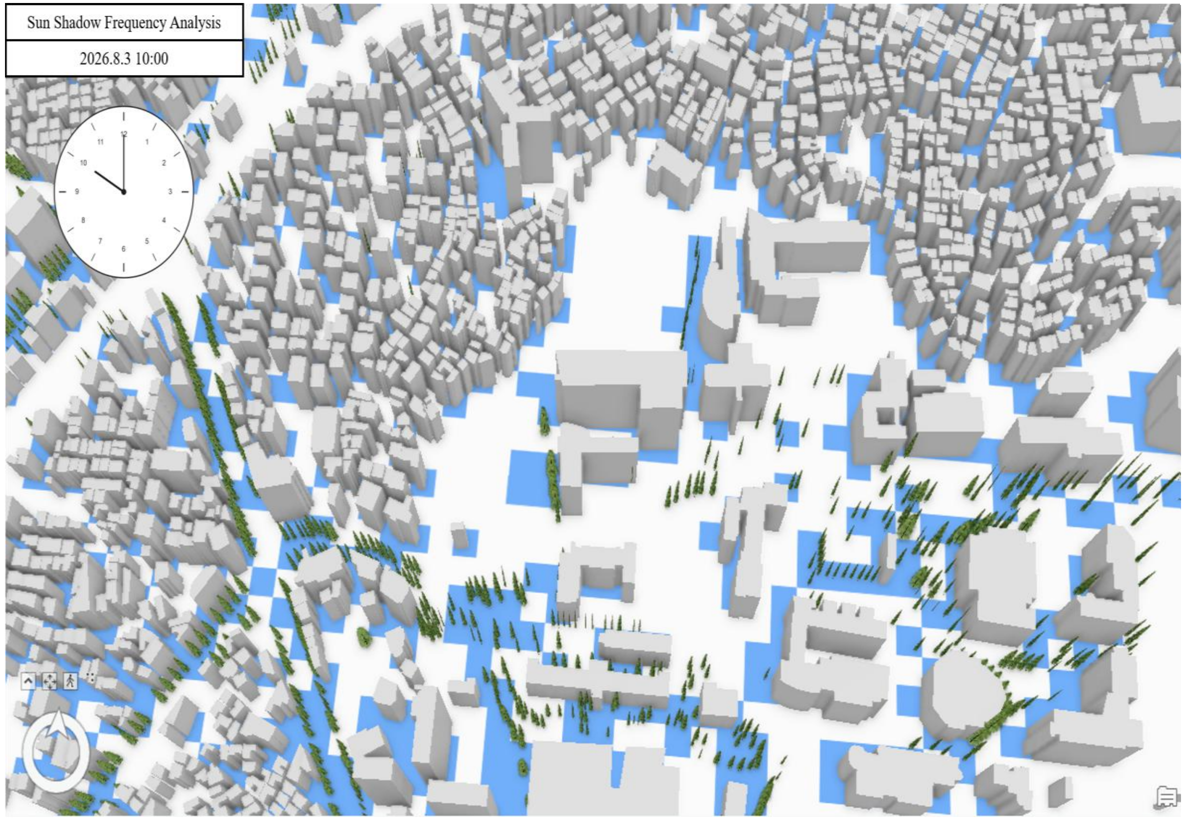
# 03 Meso-Scale Walkability Index : Sun Shadow Frequency Analysis

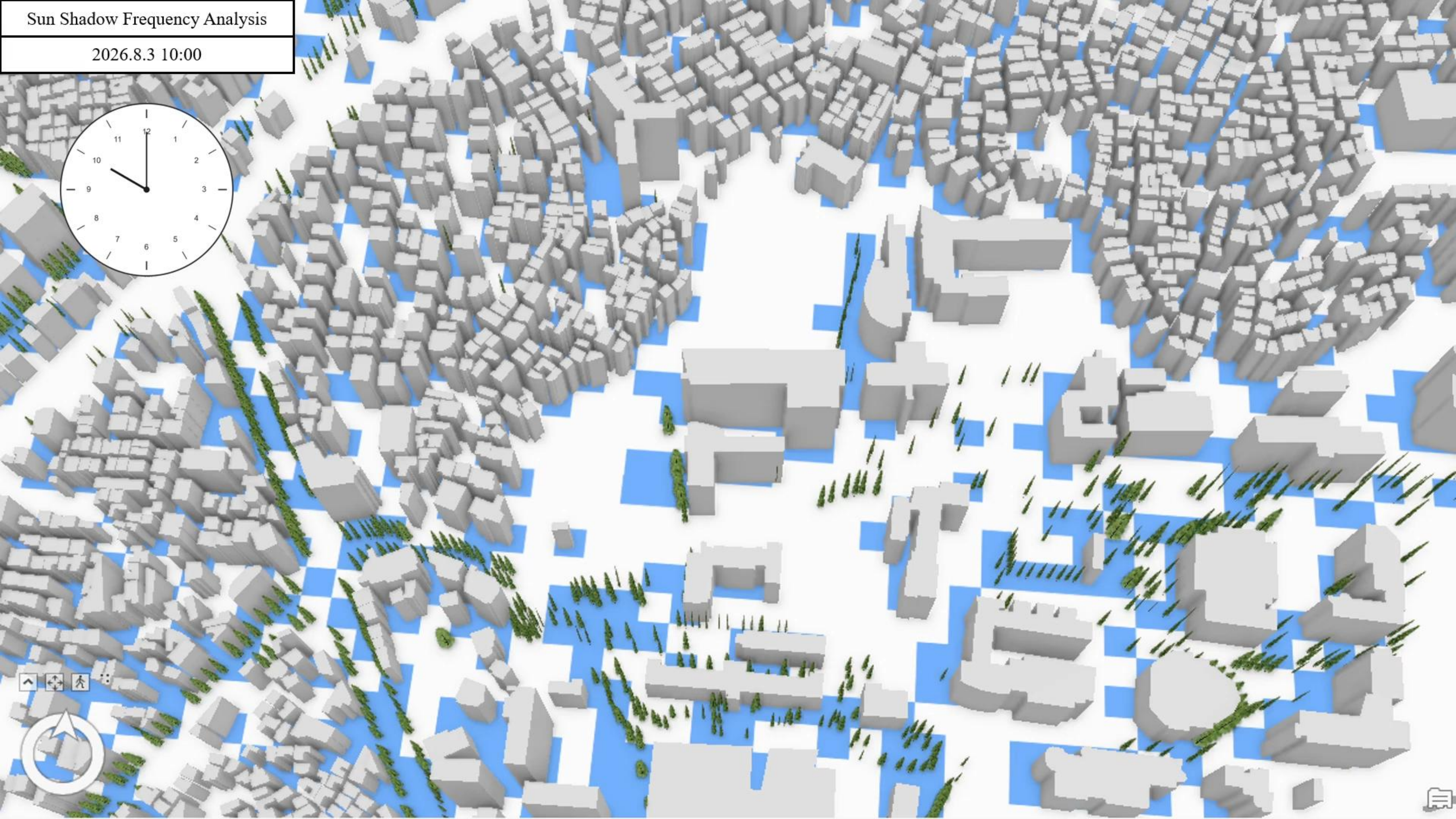
- Sun shadow frequency was analyzed using 3D building and tree models, with date, time range, interval, and cell size specified as key parameters.
- Seasonal analyses were conducted for Spring/Fall, Summer, and Winter to capture temporal variation in walkability.

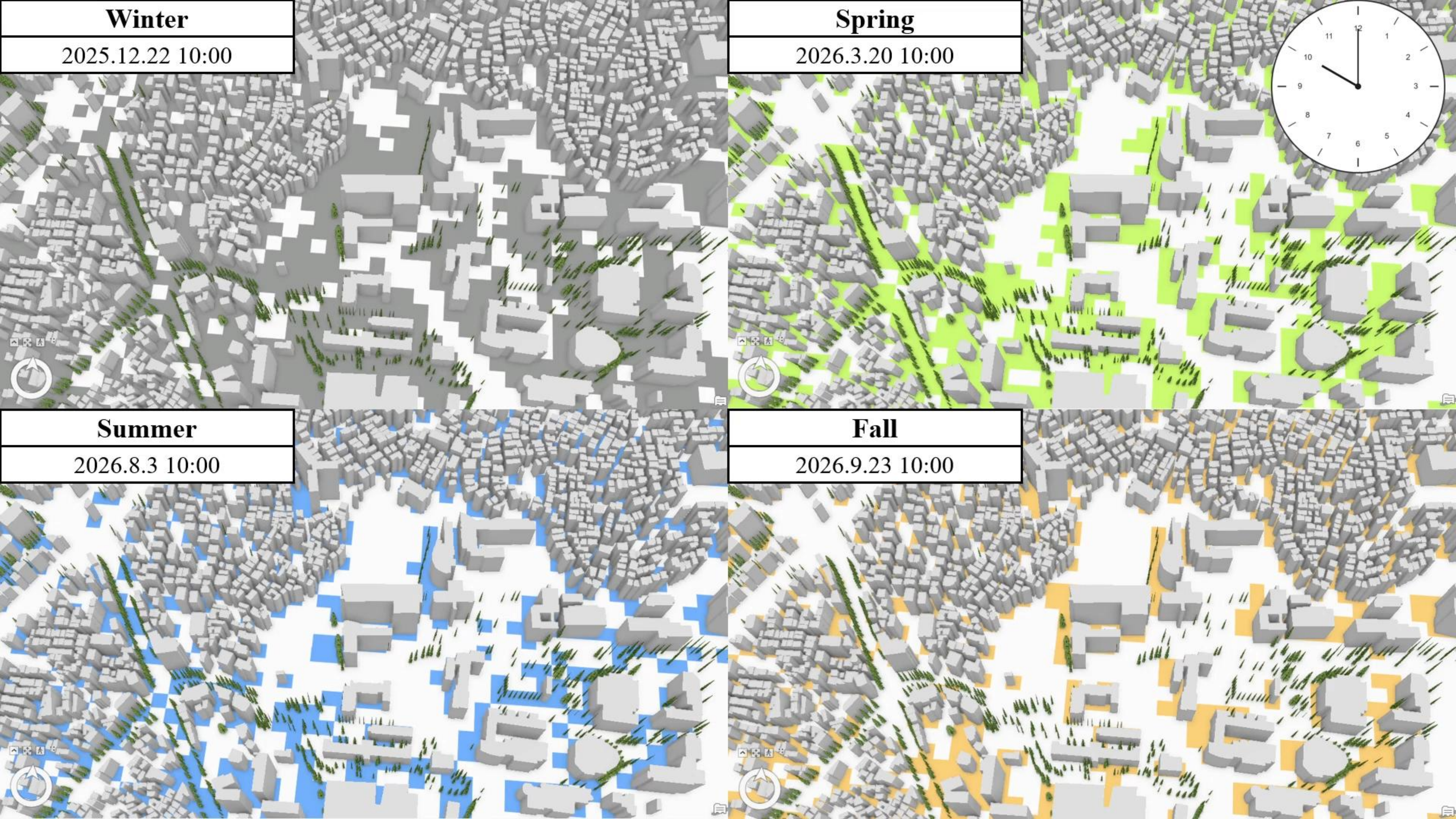
**3D Building and Tree Modelling**

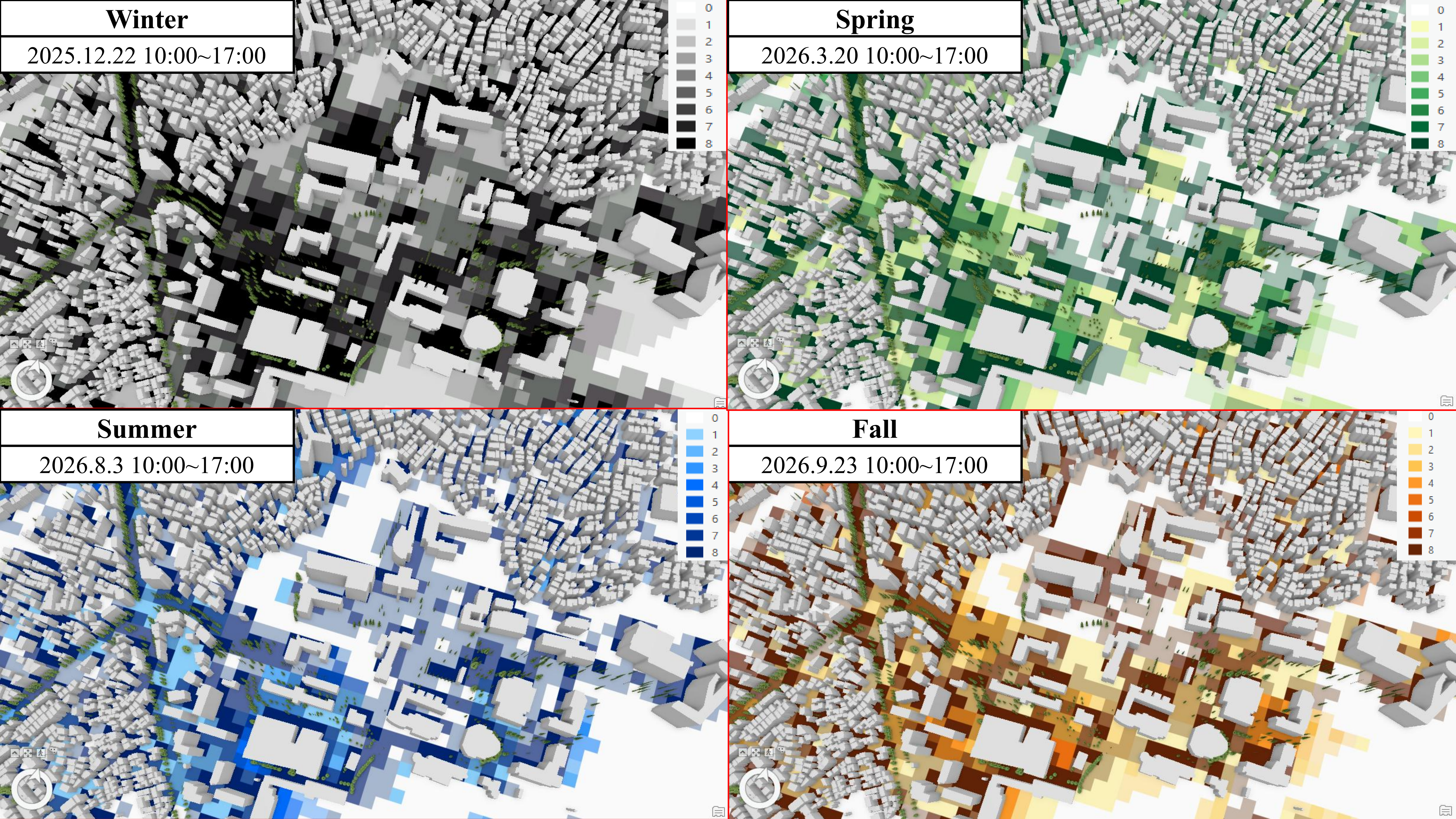



**Sun Shadow Frequency Analysis example (26.8.3)**




Sun Shadow Frequency Analysis
2026.8.3 10:00


Winter
2025.12.22 10:00
Spring
2026.3.20 10:00
12
1
2
3
4
5
6
7
8
9
10
11
Summer
2026.8.3 10:00
Fall
2026.9.23 10:00


Winter
2025.12.22 10:00~17:00
0
1
2
3
4
5
6
7
8
Spring
2026.3.20 10:00~17:00
0
1
2
3
4
5
6
7
8
Summer
2026.8.3 10:00~17:00
0
1
2
3
4
5
6
7
8
Fall
2026.9.23 10:00~17:00
0
1
2
3
4
5
6
7
8

# 03 Multiscale Walkability(3): Defining the Macro-Scale Walkability Index

- Macro-scale walkability extends beyond local pedestrian environments to capture the structural walking potential and locational attractiveness of each area within the citywide transport network.
- It is constructed using land-use mix, jobs–housing balance, and the relative travel cost of public transport versus private cars.

## Components of Macro-Scale Walkability

| | |
|---|---|
| • **Land-use mix** | Entropy index of surrounding administrative areas |
| • **Jobs–housing balance** | Ratio of workers to residents in surrounding areas |
| • **Public transport efficiency** | Public transport travel cost / car travel cost |

## Macro-Scale Walkability Index

$$W_{macro} = Z\left( ln\left( \sum_{j\neq i}\left[ \frac{(L_j + JHR_j)}{Cost_{i,j}^{PT} / Cost_{i,j}^{Auto}} \times \left( \frac{V_{i,j}}{\sum_k V_{i,k}} \right) \right] + 1 \right) \right)$$

$W_{macro}$ = macro-scale walkability

$L_j$ = land-use mix at destination j

$JHR_j$ = jobs–housing ratio at destination j

$Cost_{i,j}^{PT}$ = public transport travel cost

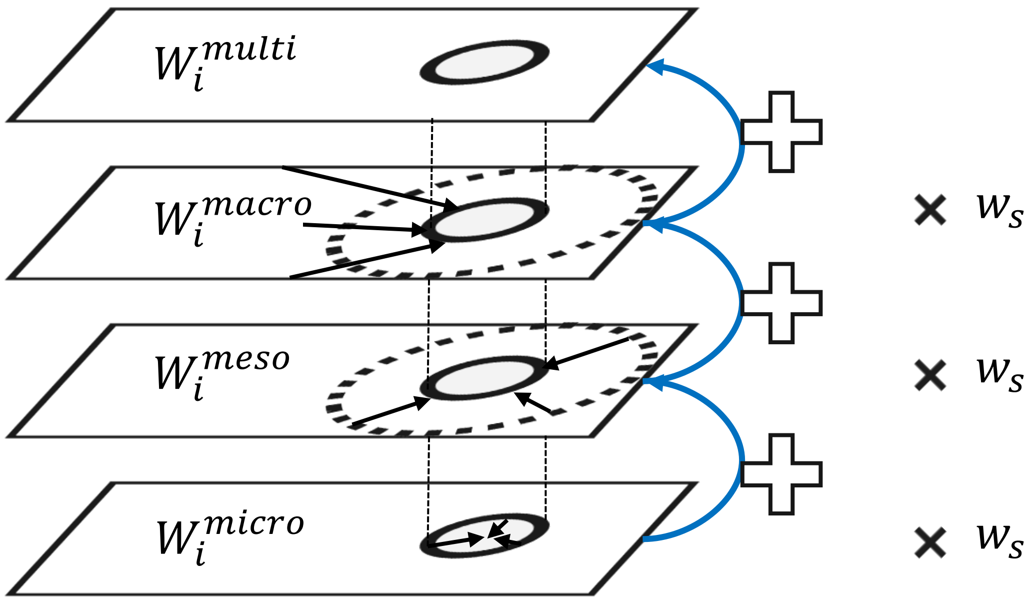
$Cost_{i,j}^{Auto}$ = private car travel cost

$V_{i,j}$ = public transport travel volume

# 03 Multiscale Walkability(4): Integrating Multiscale Walkability

- This study conceptualizes the walking environment across three spatial scales and integrates them by reflecting the functional role and spatial hierarchy of each scale.
- Both equal weighting and data-driven weighting are applied to compare alternative structures of multiscale walkability.

**Integration of Multiscale Walkability**




**Multiscale Walkability Integration: Data-Driven Weighting**

$$X_i = (W_i^{micro}, W_i^{meso}, W_i^{macro})$$

$$p_{i,s} = \frac{W_i^s}{\sum_{i-1}^{n} W_i^s}, \qquad e_s = -\frac{1}{\ln n}\sum_{i-1}^{n} p_{i,s}\ln(p_{i,s})$$

$$w_s = -\frac{1-e_s}{\sum_{s\in\{micro,meso,macro\}}(1-e_s)}$$

$$W_i^{multiscale} = \sum_{s\in\{micro,meso,macro\}} w_s \cdot W_i^s$$

$W_i^s$ = walkability of location i at scale s

$p_{i,s}$ = relative share of location i within scale s

$e_s$ = entropy of scale s (distributional uniformity)

$W_i^{multiscale}$ = multiscale walkability of location i

$w_s$ = entropy-based weight for scale s

# 03 Research Scope and Data Used

- The temporal scope is **2024**, and the spatial scope is **Seoul**. The main datasets used in this study are summarized below.

| Category | Data | Description | Source |
|---|---|---|---|
| **Micro-Scale Walkability** | Pedestrian network | • Physical conditions of the pedestrian network, such as length and location | • Seoul Open Data Plaza |
| | Elevation data | • Elevation at each point, used to calculate slope | • VWorld |
| | Crosswalks | • Crosswalk locations | • Seoul Open Data Plaza |
| | Floating population | • Used to estimate pedestrian LOS at specific locations based on Seoul urban data | • Seoul Open Data Plaza |
| **Meso-Scale Walkability** | POI data | • National Point of Interest information | • National Spatial Data Infrastructure Portal |
| | Bus stops | • Bus stop locations | • Seoul Open Data Plaza |
| | Subway stations | • Subway station locations | • Seoul Open Data Plaza |
| | Public bike stations | • Ttareungi bike-sharing station locations | • Seoul Open Data Plaza |
| | Seoul building data | • Building data in Seoul; floor information used for 3D modelling and sun shadow frequency analysis | • VWorld |
| **Macro-Scale Walkability** | Land use | • Entropy index calculated from land-use data | • Seoul Open Data Plaza |
| | Workers and residents | • Number of workers and residents by administrative dong | • Seoul Commercial District Analysis Service |
| | Transport network | • Road and public transport networks | • KTDB |
| | Mode O/D | • Passenger-car and public transport trip volumes | • KTDB |
| **Estimated Pedestrian Volume** | De facto population data | • Real-time population based on KT mobile communication data | • KT Telecom Data |
| | Daily mobility data | • Mobility data at the 250 m grid level based on KT mobile communication data | • KT Telecom Data |
| **Walking Satisfaction** | Walking satisfaction survey | • Seoul Survey responses related to walking satisfaction | • Seoul Survey |
| **Commercial Area Analysis** | Sales and transaction data | • Estimated store-level sales within Seoul commercial areas | • Seoul Credit Guarantee Foundation |

# 4. Results

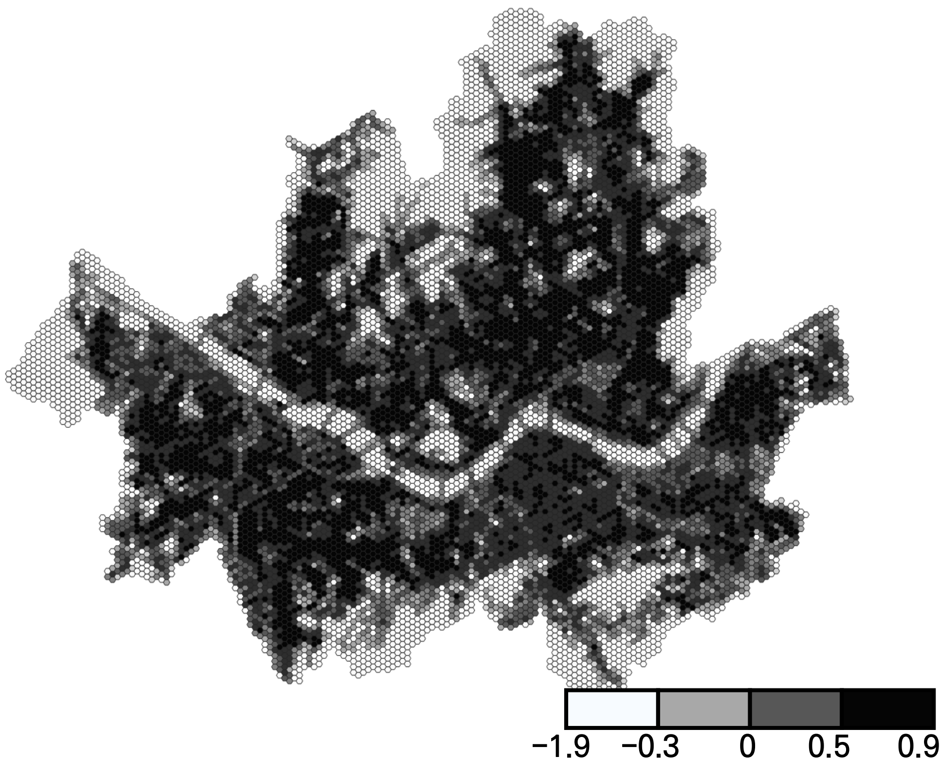
I. Construction of Multiscale Walkability

II. Market Activity Analysis Results

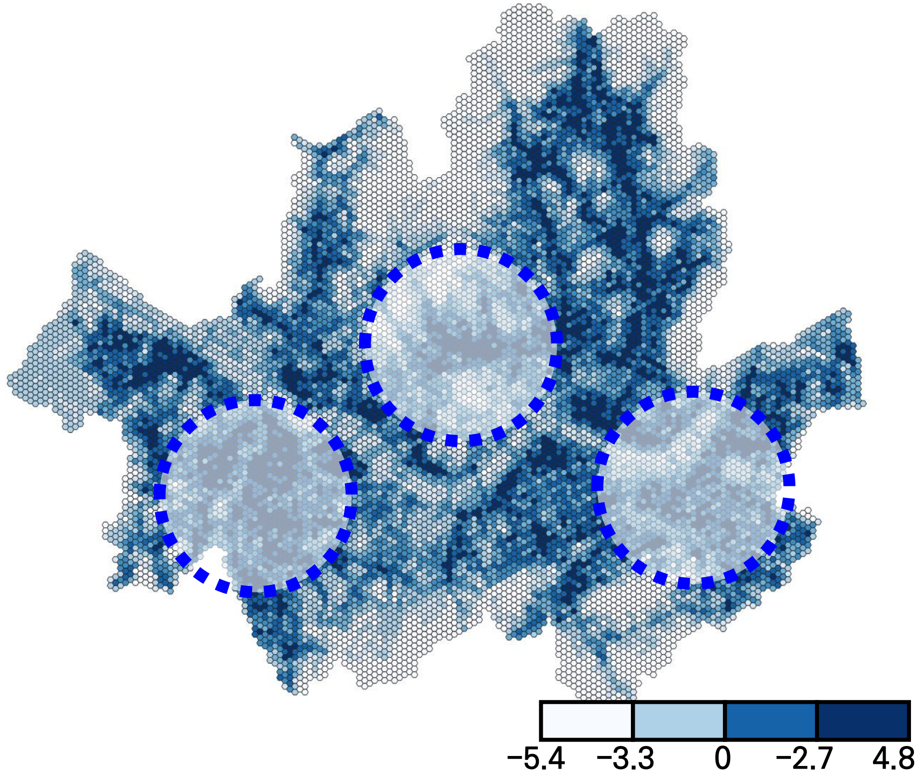
# 04 Construction of Multiscale Walkability (1): Micro-Scale Walkability index

- Pedestrian network supply is concentrated in several **central areas of Seoul.**
- Micro-scale walkability shows a more **spatially fragmented pattern**, reflecting the combined effects of sidewalk supply, slope, and crosswalk connectivity.

## Distribution of Pedestrian Network Supply (Seoul)

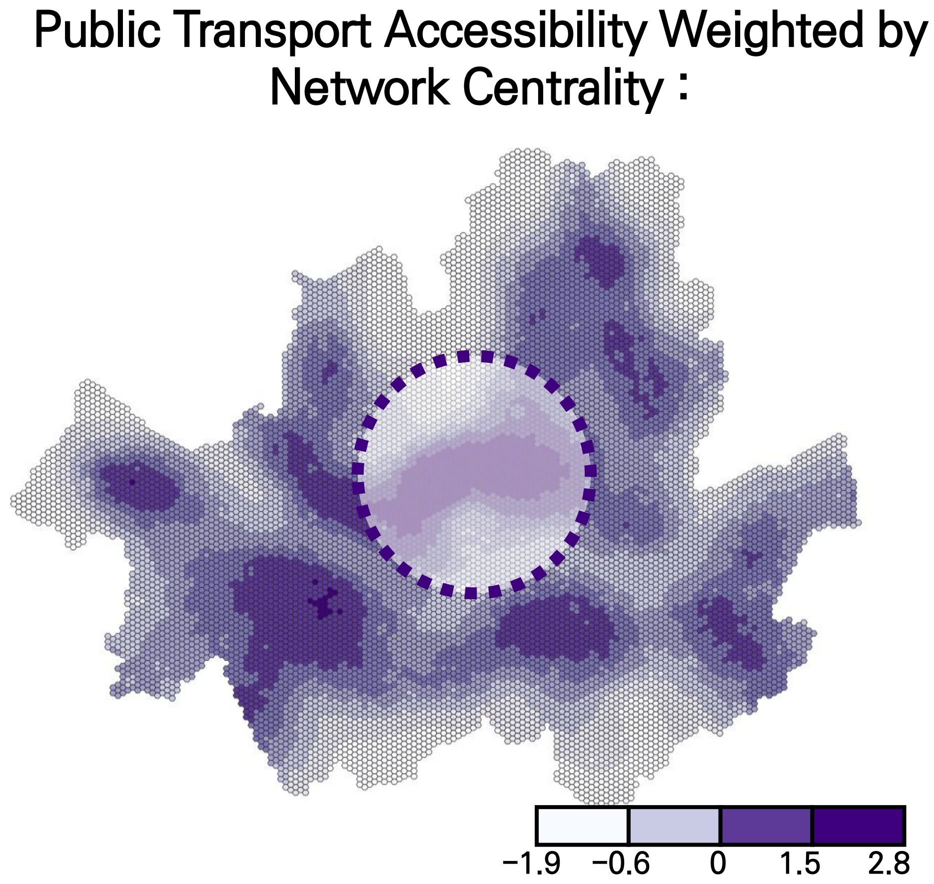


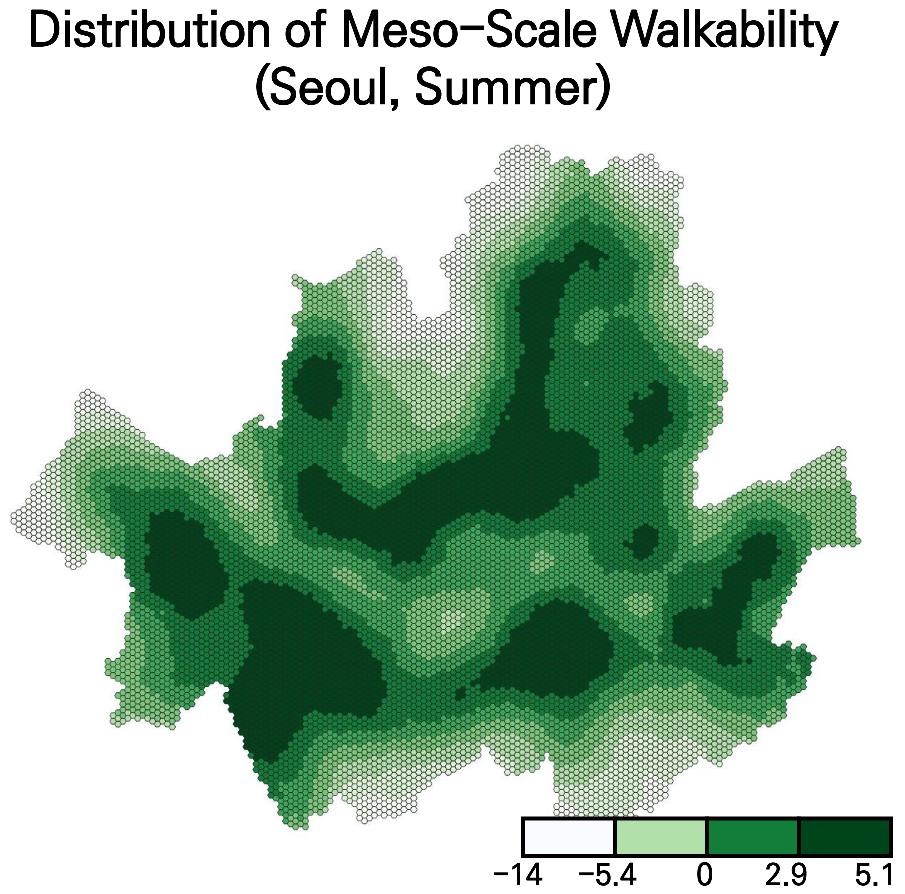

## Distribution of Micro-Scale Walkability (Seoul)



# 04 Construction of Multiscale Walkability (2): Meso-Scale Walkability index

- Meso-scale walkability captures network accessibility within a 2 km range.
- It connects micro- and macro-scale walkability through destinations, public transport, density, and seasonal comfort.

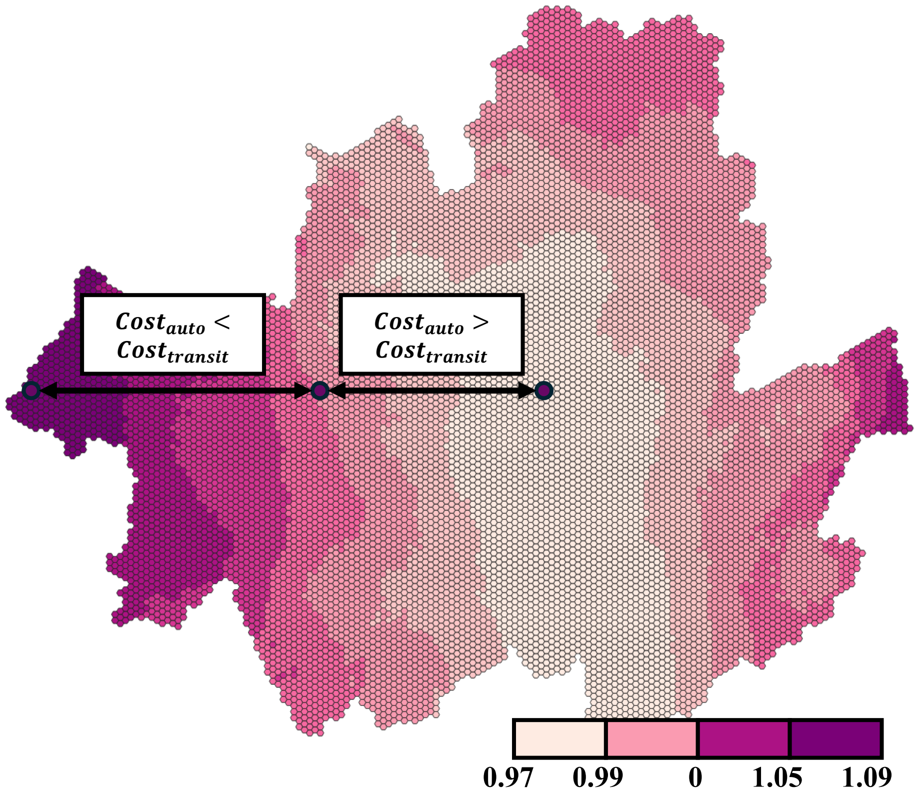


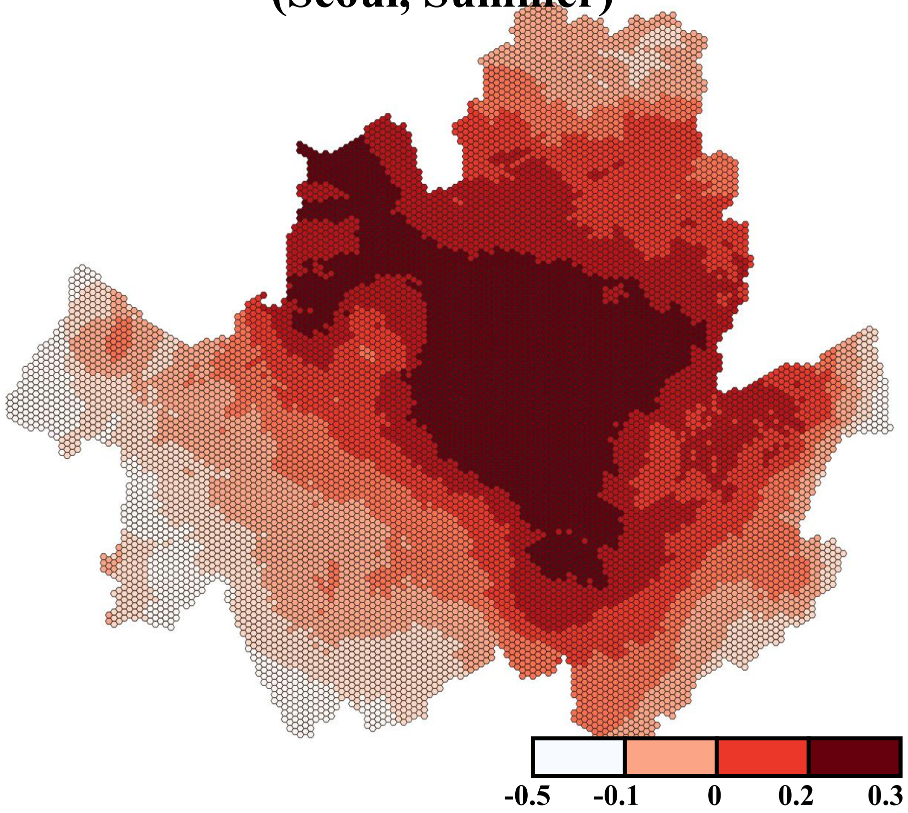
# 04 Construction of Multiscale Walkability (3): Macro-Scale Walkability index

- Public transport is relatively more competitive in central Seoul, while private cars are more competitive in peripheral areas.
- Combined with land-use mix and jobs–housing balance, macro-scale walkability forms a strong central-city pattern.

**Public Transport / Car Travel Cost Ratio (Seoul)**



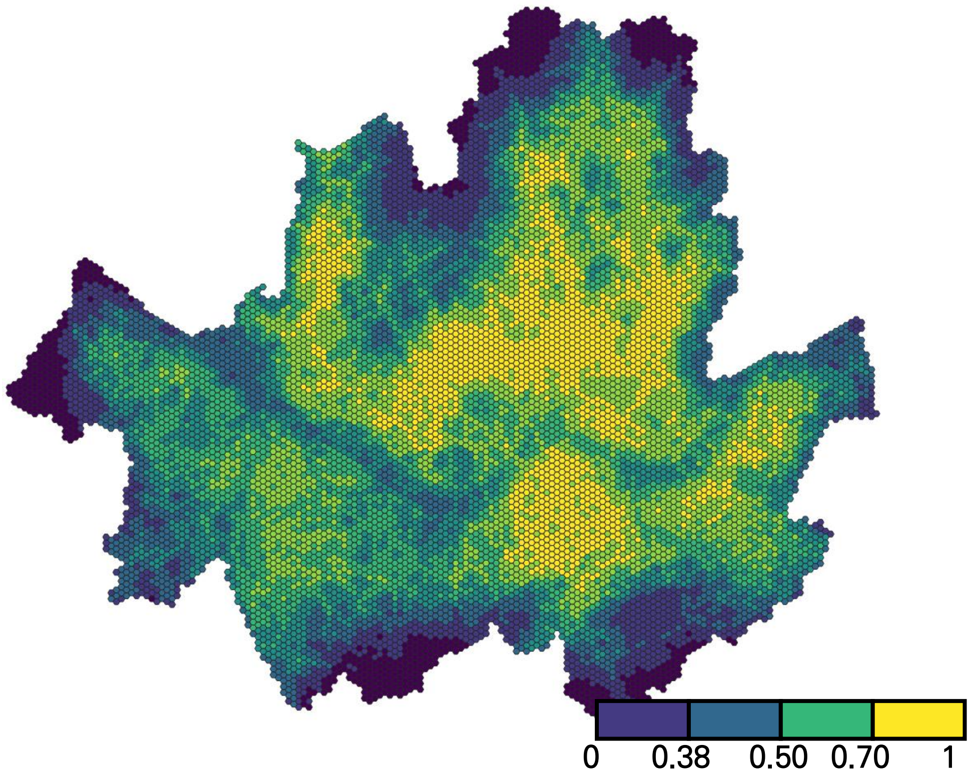

**Distribution of Macro-Scale Walkability (Seoul, Summer)**



# 04 Construction of Multiscale Walkability (4): Multiscale Walkability measurement

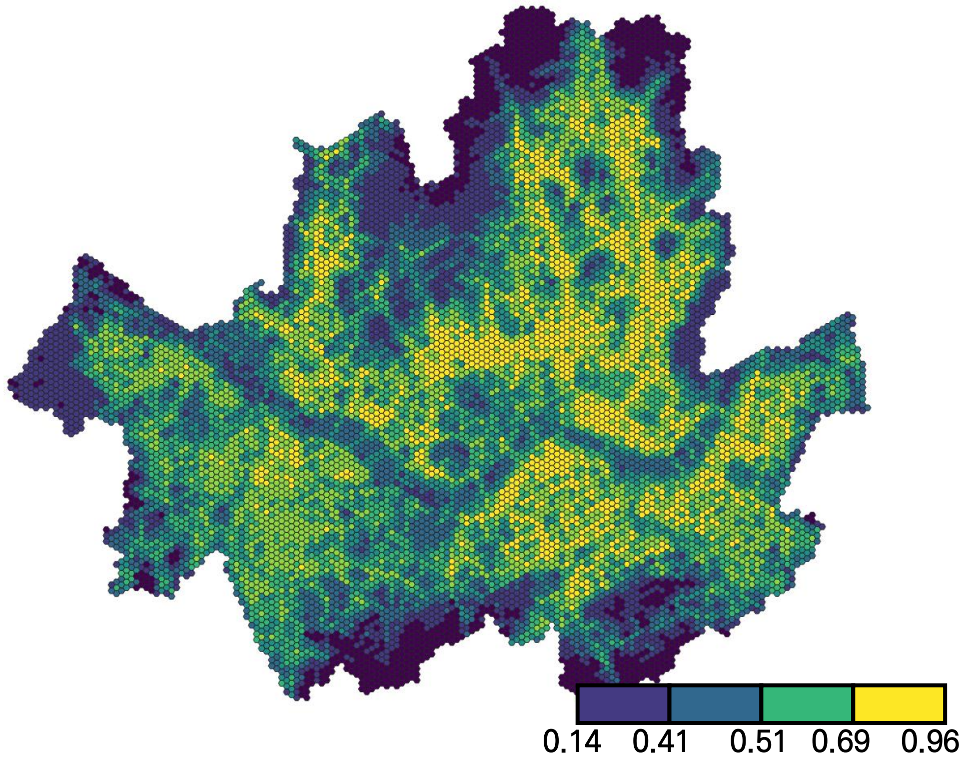
- Multiscale walkability integrates scale-specific walking environments to represent urban walkability.
- Two integration approaches were examined: equal weighting and data-driven weighting.
- Data-driven weighting reveals more distinct spatial variation across Seoul.

**멀티스케일 워커빌리티 통합 : 균일 통합방식**




**멀티스케일 워커빌리티 통합 : 정보량 반영 통합**

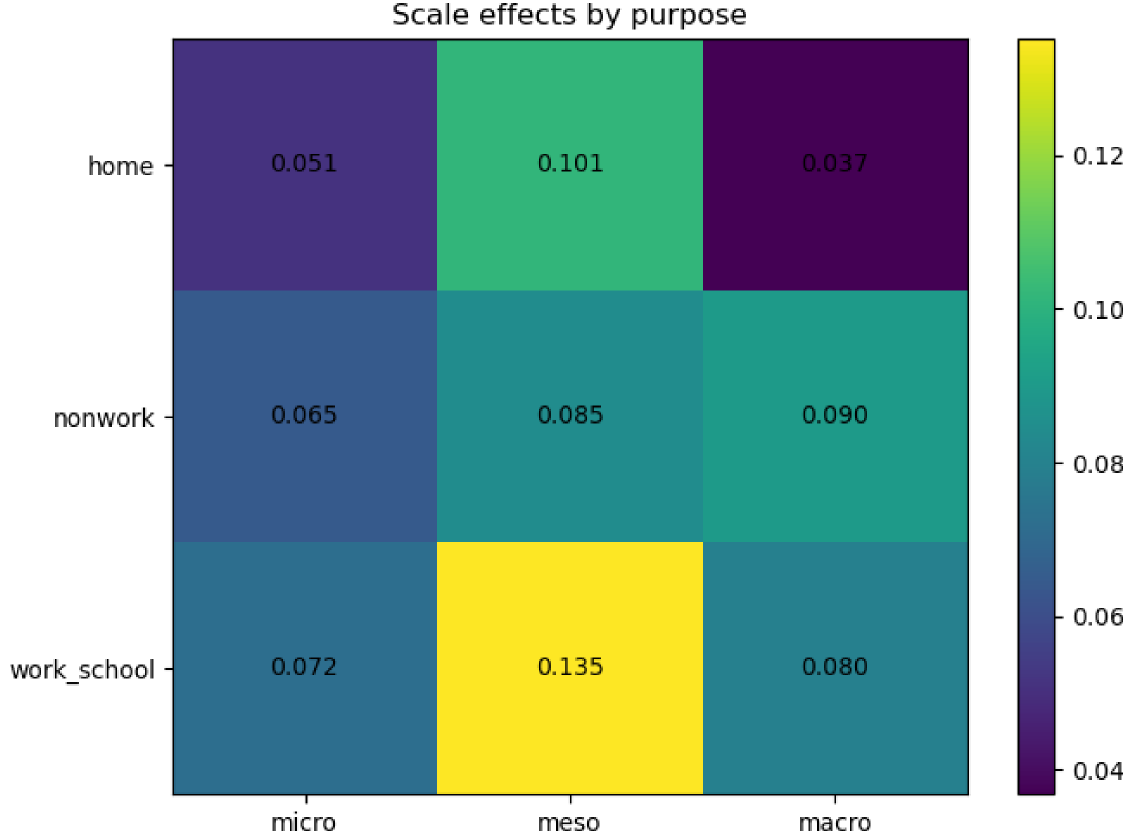
$$W_{multiscale} = 0.56 * W_{micro} + 0.19 * W_{meso} + 0.25 * W_{macro}$$



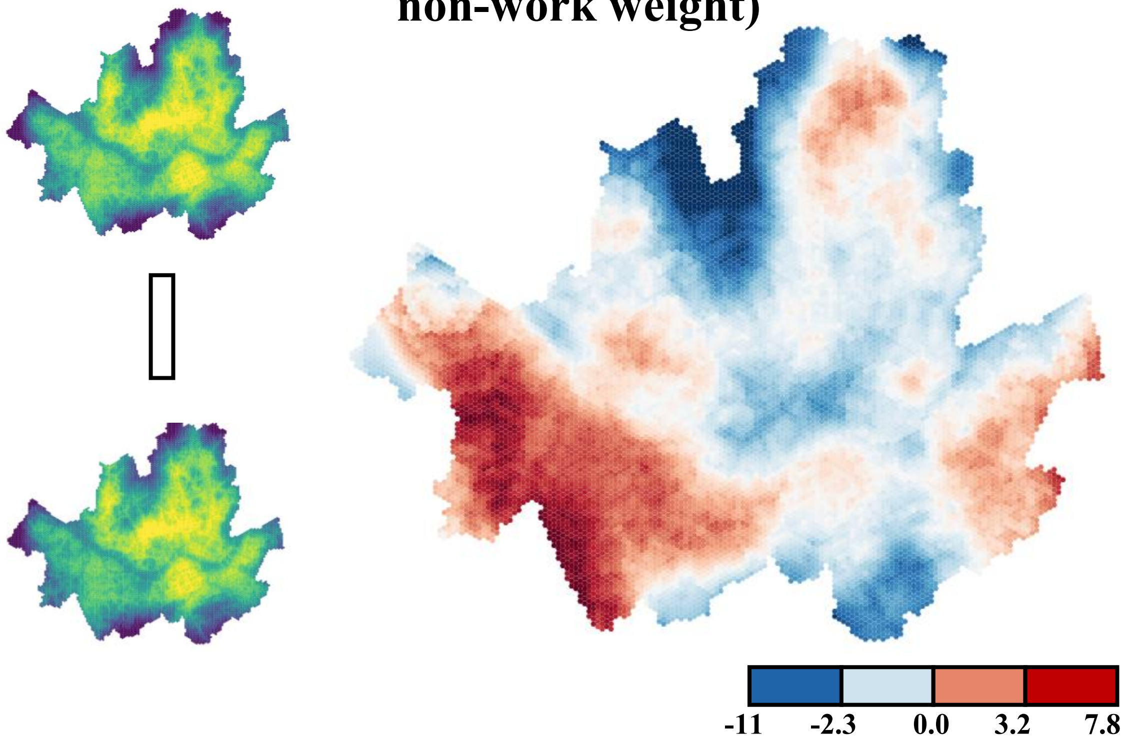
# 04 Construction of Multiscale Walkability (5): Coefficient-Based Weighting

- Walking-volume models were estimated by trip purpose, controlling for population, and standardized coefficients were used as relative weights.
- Meso-scale walkability received the largest weight, indicating the importance of destination accessibility, public transport, and network connectivity.

**Scale-Specific Coefficients by Trip Purpose**



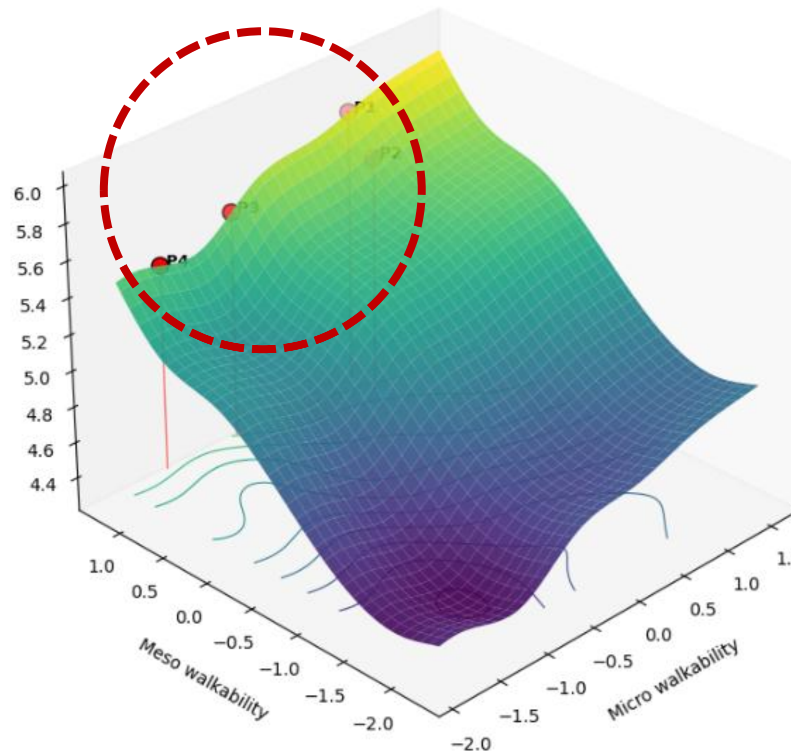

**Difference in Weighted Multiscale Walkability: Commuting vs. Non-Work Trips (Red: higher commuting weight / Blue: higher non-work weight)**



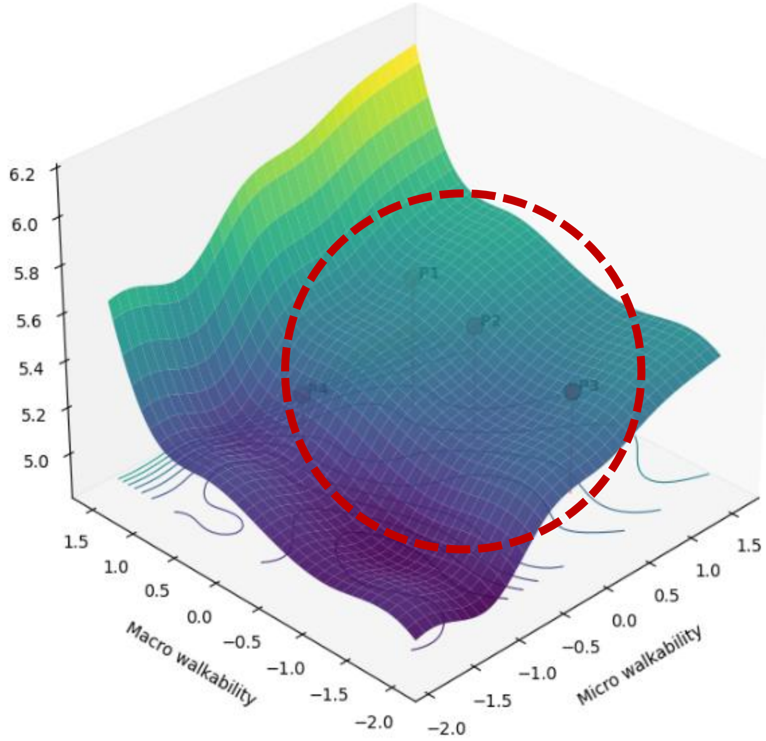
# 04 Construction of Multiscale Walkability (6): Complementary Validation Using Walking Volume

- GAM(Generalized Additive Model) was applied to examine nonlinear relationships and threshold effects between scale-specific walkability and walking volume.
- The meso scale showed positive threshold effects, with a marked increase in log walking volume when meso-scale walkability reached approximately 0.946–1.225.

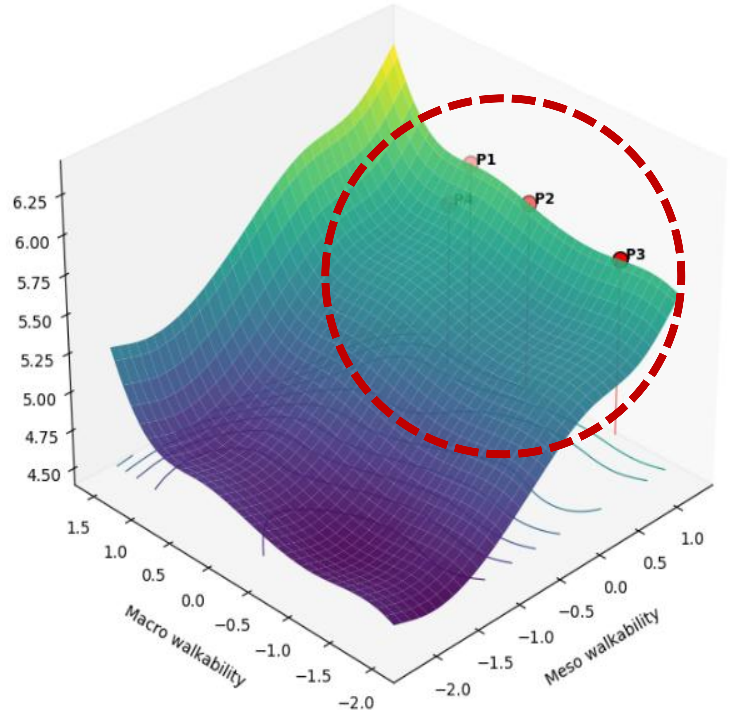
**Micro–Meso Response of Walking Volume**




**Micro–Macro Response of Walking Volume**




**Meso–Macro Response of Walking Volume**



# 04 Application of Multiscale Walkability: Example of Walking and Accessibility Benefits

- Validated walkability effects can be translated into equivalent walking-time benefits.
- Equivalent time is estimated by comparing walkability coefficients with the walking-time coefficient.
- These values can support interpretation of time savings and accessibility benefits within existing appraisal frameworks.

## Example of Equivalent-Time Effects by Trip Purpose

Equivalent-Time(ET) Coefficient

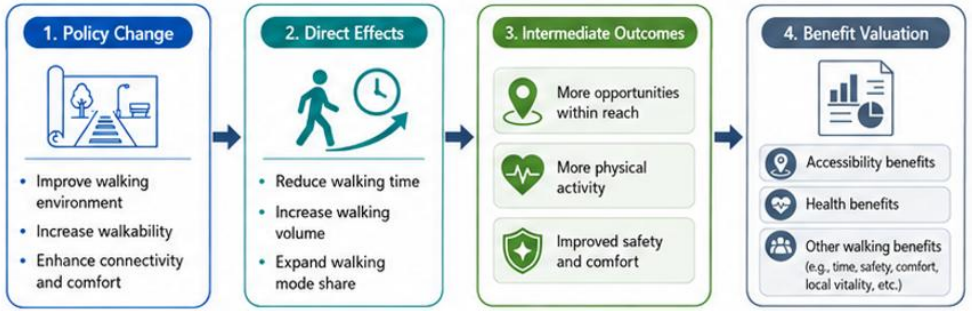
$$\log(1+walk) = \beta_0 + \beta_{micro} W_{micro} + \beta_{eso} W_{meso} + \beta_{macro} W_{macro} + \beta_p \log(1+Pop) + \beta_T T_{net}$$

$$ET = \left|\frac{\beta_s}{\beta_t}\right|,\ s \in \{micro, meso, macro)$$

### Example of Equivalent-Time Coefficient by Trip Purpose

| Category | | Coefficient | Equivalent Time (min) | Category | | Coefficient | Equivalent Time (min) |
|---|---|---|---|---|---|---|---|
| Commuting | Micro | 0.056 | 6.55 | Non-work | Micro | 0.056 | 3.67 |
| | Meso | 0.083 | 9.68 | | Meso | 0.048 | 3.22 |

### Example of Benefit Appraisal Framework



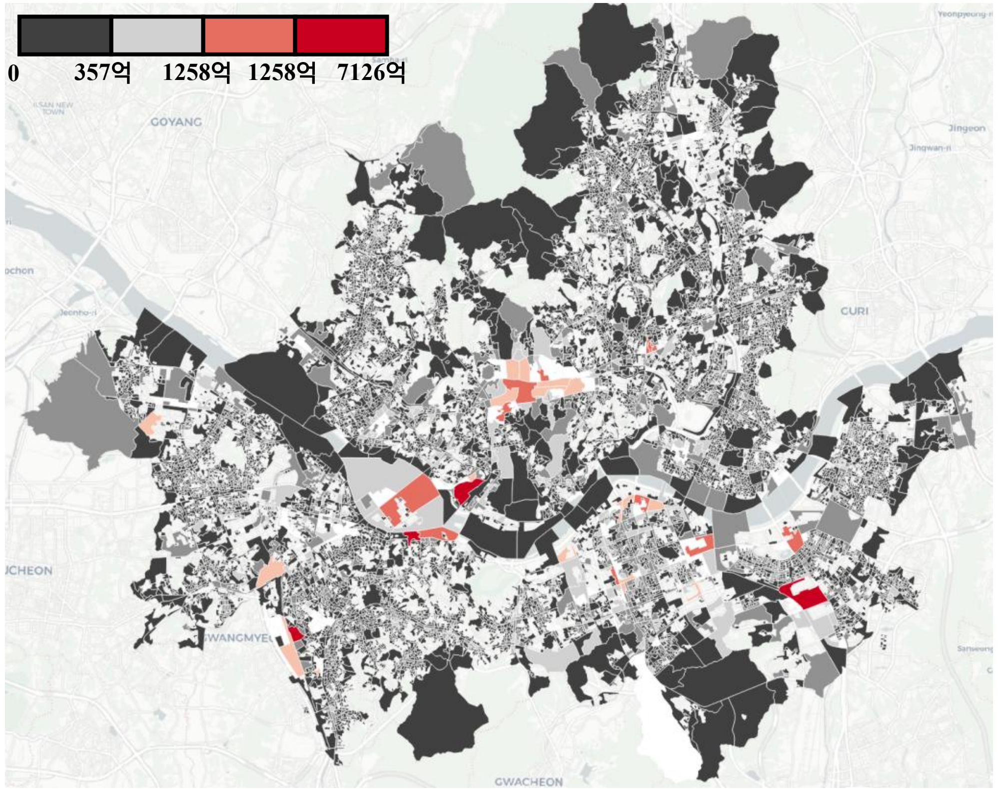

### Examples of International Appraisal Guidelines

| Source | Key Guidance |
|---|---|
| OECD / ITF | Walking and cycling quality, time value, health benefits |
| WHO HEAT | Health economic valuation of physical activity |
| NZ MBCM | Monetised benefits of active transport |
| OECD/ITF | Transport benefit appraisal framework |

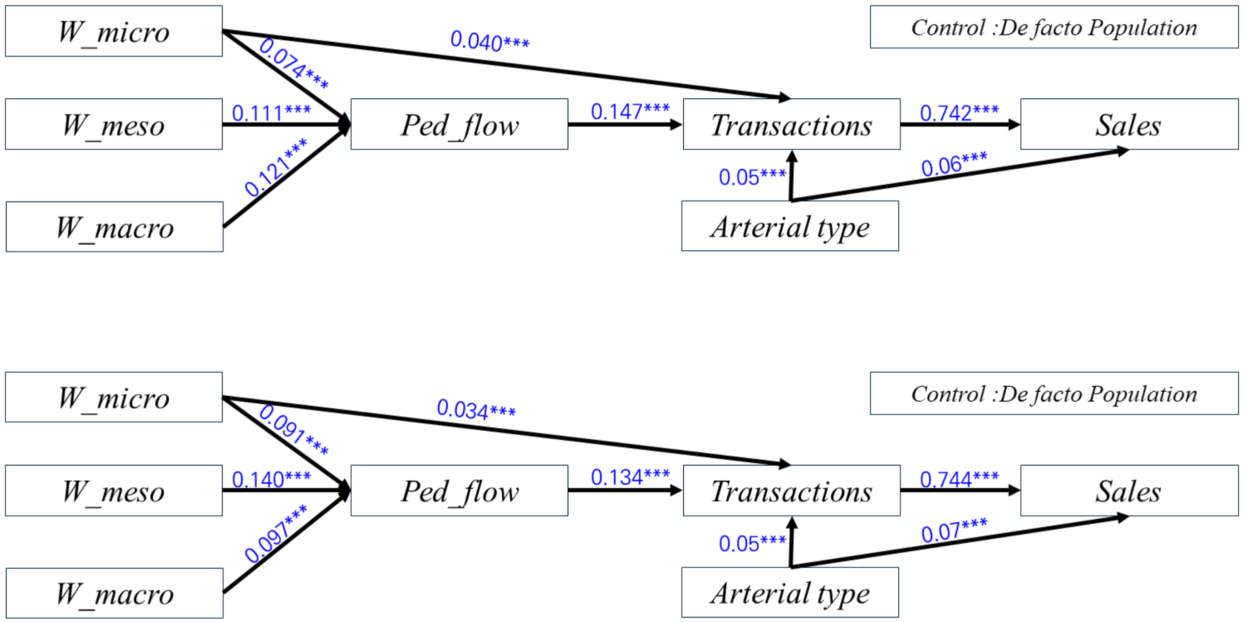
# 04 Market Activity Analysis Results (1): Multiscale Walkability and Commercial Activity

- Market activity shows spatial clustering, indicating spatial dependence across commercial areas.
- Structural Equation Modeling (SEM) results show that multiscale walkability is **positively associated with pedestrian flow, which is linked to transactions and sales.**
- Some scales also show **direct relationships with commercial activity**, suggesting both direct and indirect pathways.

**Distribution of Commercial Activity (Quarterly Sales)**



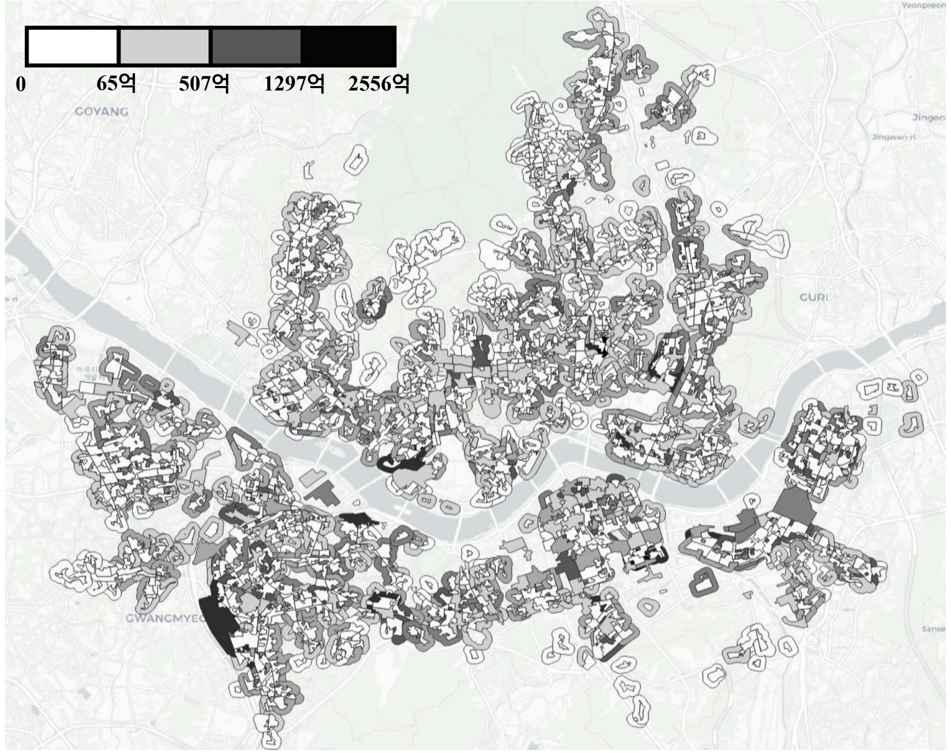

**Structural Equation Modeling (SEM) Results by Season**
(Summer vs. Winter)



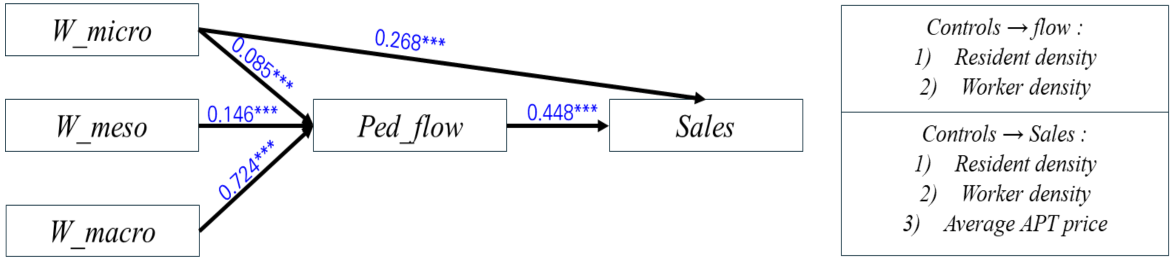
# 04 Market Activity Analysis Results (2): Commercial-Area-Level Analysis of Multiscale Walkability

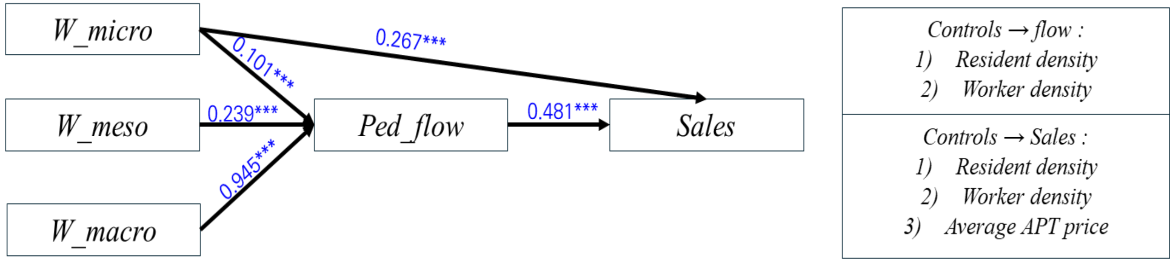
- SEM results show that multiscale walkability is **positively associated with pedestrian flow and sales** in both summer and winter.
- Pedestrian flow also shows a strong positive relationship with sales, suggesting an indirect pathway from walkability to market activity.

**Distribution of Average Weekday Sales by Commercial Area**




Structural Equation Modeling (SEM) Results by Season: Summer vs. Winter



# 04 Market Activity Analysis Results (4): Policy Translation of Multiscale Walkability Improvements

- Physical and accessibility improvements were converted into standardized changes in scale-specific walkability indicators and linked to SEM path coefficients.
- For micro-scale scenarios, improvements in sidewalks, slopes, and crosswalks were translated into potential changes in sales and transactions.

## Common Policy Translation Framework for Multiscale Walkability

Policy Intervention

↓

Change in Walkability

↓

SEM Path Coefficients

↓

Policy Impact

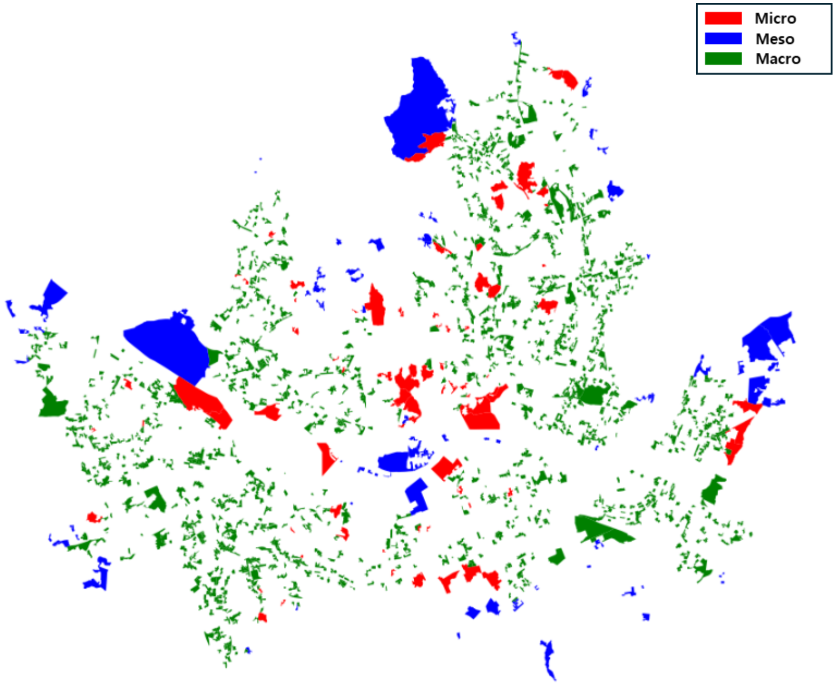
## Micro-Scale Walkability Improvement Scenarios and Estimated Effects


| Scenario | | Estimated effect |
|---|---|---|
| Sidewalk +30 m, Crosswalk +1 | $\Delta Z(W_{micro})$ | Sales +8.02% / Transactions +4.75% |
| Slope −2%, Crosswalk +1 | $\Delta Z(W_{micro})$ | Sales +6.01% / Transactions +3.59% |
| Sidewalk +50 m, Slope −2%, Crosswalk +1 | $\Delta Z(W_{micro})$ | Sales +15.04% / Transactions +8.81% |

# 04 Market Activity Analysis Results (4): Policy Translation of Multiscale Walkability Based on Nonlinear Walking Responses

- **GAM-based nonlinear walking responses by walkability scale were linked with SEM-based structural relationships between pedestrian flow and sales to examine potential commercial impacts of localized walking-volume responses.**

## Scale-Specific Vulnerable Areas of Walkability Indicators (Bottom 1–2 Quantiles)




## Micro-Scale Walkability Improvement

Nonlinear Walking Response

SEM Relationships and Path Effects

Market-Impact Sensitivity

## Meso- and Macro-Scale Walkability Improvement

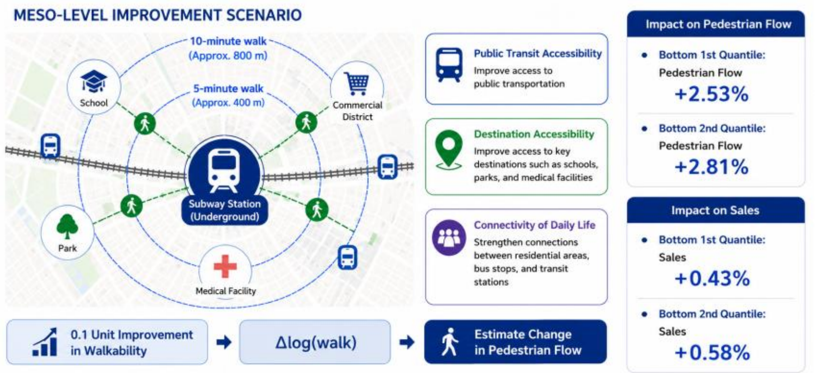
Nonlinear Walking Response

Local Slope of Walking Response

Estimate Walking-Volume Change

## Improvement Scenario and Estimated Effects (Increase Wmicro by 0.1)


## Improvement Scenario and Estimated Effects (Increase Wmeso and Wmacro by 0.1)



# 5. Conclusion

I. Conclusions and Limitations

II. Future Research

UNIVERSITY OF SEOUL

# 05 Conclusions and Future Research

## • Conclusions and Limitations

1. Walkability is conceptualized as a **multiscale framework** consisting of micro, meso, and macro dimensions.
2. Multiscale walkability is **structurally linked to market activity**, with different effects across spatial scales.
3. The study identifies pathways from **walkability to pedestrian flow and sales**, and demonstrates potential policy translation of walkability improvements.
4. **However, the analysis is observational, limiting causal interpretation, and spatial heterogeneity may remain.**

## • Future Research

1. Expand behavioral data and conduct panel or simulation-based analyses to **evaluate the magnitude and persistence of walkability effects**.
2. Extend multiscale walkability to **mobility planning and multimodal environments**, including emerging mobility systems.
3. Apply the framework to **street redesign and policy evaluation** to support pedestrian-oriented urban planning.

# References


Khavarian-Garmsir, A. R., Sharifi, A., & Sadeghi, A. (2023). The 15-minute city: Urban planning and design efforts toward creating sustainable neighborhoods. Cities, 132, 104101.

Aparicio, J. T., Arsenio, E., Santos, F. C., & Henriques, R. (2024). Walkability defined neighborhoods for sustainable cities. Cities, 149, 104944.

He, X., & He, S. Y. (2023). Using open data and deep learning to explore walkability in Shenzhen, China. Transportation research part D: transport and environment, 118, 103696.

Wu, Y., Liu, Q., Hang, T., Yang, Y., Wang, Y., & Cao, L. (2024). Integrating restorative perception into urban street planning: A framework using street view images, deep learning, and space syntax. Cities, 147, 104791.

Forsyth, A. (2015). What is a walkable place? The walkability debate in urban design. Urban design international, 20(4), 274-292.

Baobeid, A., Koç, M., & Al-Ghamdi, S. G. (2021). Walkability and its relationships with health, sustainability, and livability: elements of physical environment and evaluation frameworks. Frontiers in built environment, 7, 721218.

Sirjani, A. H., & Szabó, Á. (2023). Evaluating walkability in Budapest through space syntax analysis: a case study of Egyetem square and Corvin promenade. Periodica Polytechnica Architecture, 54(3), 238-246.

Ros-McDonnell, D., de-la-Fuente-Aragón, M. V., Ros-McDonnell, L., & Cardós, M. (2024). Toward resilient urban design: Pedestrians as an important element of city design. Urban Science, 8(2), 65.

Sadeghi, O., & Serugendo, G. D. M. (2024). Assessing walkability in urban environments: A comprehensive overview.

Rafiemanzelat, R., Emadi, M. I., & Kamali, A. J. (2017). City sustainability: the influence of walkability on built environments. Transportation research procedia, 24, 97-104.

Yu, X., Hoggenmüller, M., Tran, T. T. M., Wang, Y., & Tomitsch, M. (2024). Understanding the interaction between delivery robots and other road and sidewalk users: A study of user-generated online videos. ACM Transactions on Human-Robot Interaction, 13(4), 1-32.

Piatkowski, D. P. (2021). Autonomous shuttles: what do users expect and how will they use them?. Journal of Urban Technology, 28(3-4), 97-115.

Chaturvedi, M., & Srivastava, S. (2023, September). Modelling and Analysis of Walk and Intelligent Feeder Service Based Last Mile Connectivity to Metro Stations. In 2023 IEEE 26th International Conference on Intelligent Transportation Systems (ITSC) (pp. 3437-3442). IEEE.

Shay, E., Khattak, A. J., & Wali, B. (2018). Walkability in the connected and automated vehicle era: a US perspective on research needs. Transportation research record, 2672(35), 118-128.

Planning, U. (2019). Walking the First/Last Mile to/from Transit: Placemaking a Key Determinant.

Liu, R., Fagnant, D. J., & Zhang, W. B. (2016). Beyond single occupancy vehicles: Automated transit and shared mobility. In Road Vehicle Automation 3 (pp. 259-275). Cham: Springer International Publishing.

Southworth, M. (2005). Designing the walkable city. Journal of urban planning and development, 131(4), 246-257.

Park, S., Deakin, E., & Jang, K. (2015). Can good walkability expand the size of transit-oriented developments?. Transportation Research Record, 2519(1), 157-164.

Pilgram, C. A., & West, S. E. (2023). Transit station area walkability. Journal of Transport and Land Use, 16(1), 267-292.

Hafiz, D., AlKhafagy, M., & Zohdy, I. (2025). Autonomous Public Transport: Evolution, Benefits, and Challenges in the Future of Urban Mobility. World Electric Vehicle Journal, 16(9), 482.

Kang, C. D. (2018). The S+ 5Ds: Spatial access to pedestrian environments and walking in Seoul, Korea. Cities, 77, 130-141.

Cervero, R., & Kockelman, K. (1997). Travel demand and the 3Ds: Density, diversity, and design. Transportation Research Part D, 2, 199–219.

Frank, L. D., et al. (2005). Linking objectively measured physical activity with objectively measured urban form: Findings from SMARTRAQ. American Journal of Preventive Medicine

Frank, L. D., et al. (2010). The development of a walkability index: Application to the Neighborhood Quality of Life Study.

Ewing, R., & Cervero, R. (2010). Travel and the Built Environment: A Meta-Analysis. Journal of the American Planning Association.

Cervero, R., & Kockelman, K. (1997). Travel demand and the 3Ds: Density, diversity, and design. Transportation Research Part D: Transport and Environment, 2, 199–219.

Watson, K. B., Whitfield, G. P., Thomas, J. V., Berrigan, D., Fulton, J. E., & Carlson, S. A. (2020). Associations between the national walkability index and walking among US adults—National Health Interview Survey, 2015. Preventive medicine, 137, 106122.

Lam, T. M., Wang, Z., Vaartjes, I., Karssenberg, D., Ettema, D., Helbich, M., ... & Lakerveld, J. (2022). Development of an objectively measured walkability index for the Netherlands. International Journal of Behavioral Nutrition and Physical Activity, 19(1), 50.


Watson, K. B., Whitfield, G. P., Thomas, J. V., Berrigan, D., Fulton, J. E., & Carlson, S. A. (2020). Associations between the national walkability index and walking among US adults—National Health Interview Survey, 2015. Preventive medicine, 137, 106122.

Lam, T. M., Wang, Z., Vaartjes, I., Karssenberg, D., Ettema, D., Helbich, M., ... & Lakerveld, J. (2022). Development of an objectively measured walkability index for the Netherlands. International Journal of Behavioral Nutrition and Physical Activity, 19(1), 50.


Frank, L. D., & Wali, B. (2024). Monitoring changes in walkability over time: An environmental exposure change detection framework with implications for equity and social justice. Sustainable Cities and Society, 117, 105808.

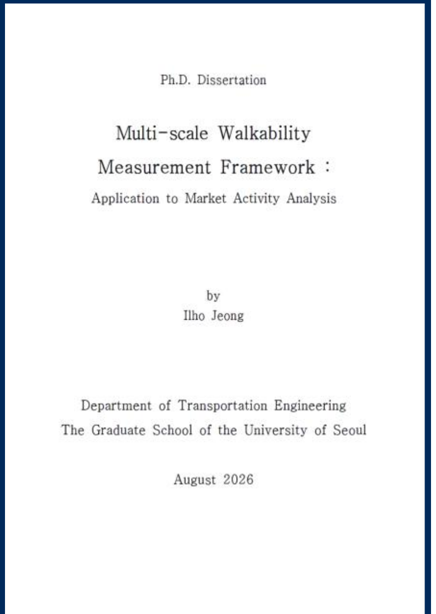
Zaleckis, K., Chmielewski, S., Kamičaitytė, J., Grazuleviciute-Vileniske, I., & Lipińska, H. (2022). Walkability compass—a space syntax solution for comparative studies. Sustainability, 14(4), 2033.

Şahin Körmeçli, P. (2023). Analysis of walkable street networks by using the space syntax and GIS techniques: a case study of Çankırı city. ISPRS International Journal of Geo-Information, 12(6), 216.


UNIVERSITY OF SEOUL

Ilho Joeng
Department of Transportation Engineering
University of Seoul
dlfgh4523@uos.ac.kr

# THANK YOU


Ph.D. Dissertation

Multi-scale Walkability
Measurement Framework :
Application to Market Activity Analysis

by
Ilho Jeong

Department of Transportation Engineering
The Graduate School of the University of Seoul

August 2026